


# Professional SCRUM MASTER

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Scrum.org
Helping people and teams solve complex problems.



*"If you haven't found it yet, keep looking. Don't settle. As with all matters of the heart, you'll know when you find it."*

*- Steve Jobs*

1

# Introductions

# Why Are You in This Class?

- Introduce yourself
- Have you used Scrum before?
- Are you a Scrum Master?
- What is your professional background?





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Make roughly even-sized teams of 5 members, or less, with each team having mixed ranges of Scrum skills and experience.

Organize your working environment.

Post for all to see:

- What is Scrum
- The purpose of a Scrum Master
- 3 things you want to learn in this class

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Exercise

# It's Your Experience. Own It.

10 minutes

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Scrum Values are the foundation for practices and behavior on a Scrum Team. Prepare a poster with a set of guidelines for us to use during this class to ensure we remain aligned with the Scrum Values.

Consider how you would like the class to operate, making it clear how it will run.




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# Professional Scrum Master Course

## PURPOSE

- Provide experience and insights so students understand how to best use Scrum to build complex products.
- Understand the theory and principles behind Scrum that guide decision making, and the Scrum Master accountability in doing so.

## AUDIENCE

- People looking to broaden and deepen their understanding of the Scrum framework and the accountability of the Scrum Master.
- Ideally have read the Scrum Guide complemented with practical experience.


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# Professional Scrum at Scrum.org

www.scrum.org/courses




Scrum Team Members
Agile Leaders
Stakeholders




Scrum Masters
Scrum Team Members




Product Owners
Scrum Masters




Agile Leaders
Scrum Masters
Product Owners




Scrum Team Members
*Developing Software*




Experienced Scrum Masters




Experienced Product Owners




Experienced Practitioners
Product Owners
Agile Leaders




Scrum Masters
Scrum Team Members
Agile Leaders




Scrum Masters
Scrum Team Members




Product Owners
Scrum Masters
UX Professionals




Scrum Masters
Scrum Team Members
Agile Leaders




Product Owners
Scrum Masters



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# Agenda

- Introductions
- Theory & First Principles
- The Scrum Framework
- Done
- Product Delivery with Scrum
- People & Teams
- The Scrum Master
- Closing

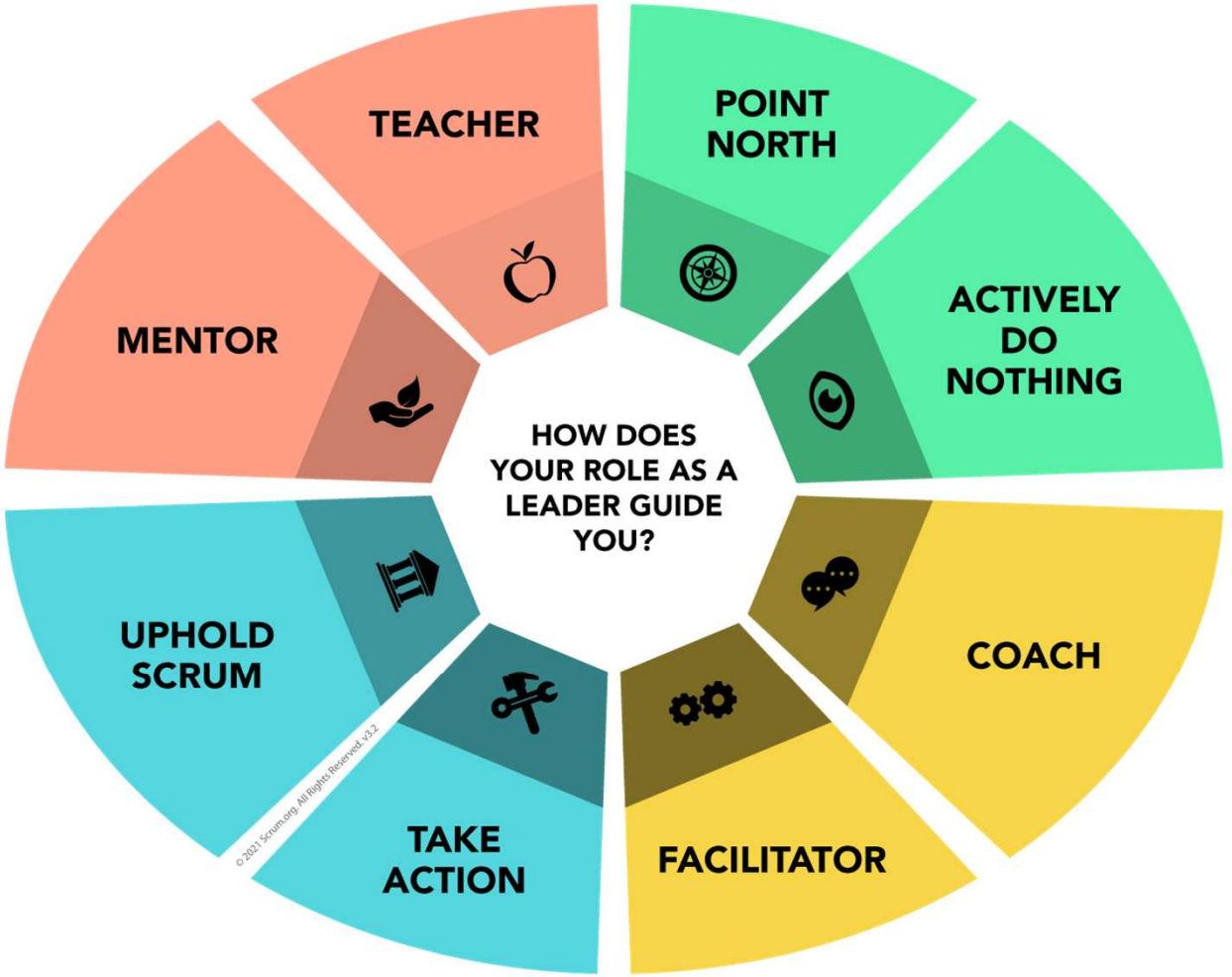
With joyful exercises along the way!


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There is no methodology for a Scrum Master to follow, yet there are intervention choices and behavioral stances from which to choose depending on context.

Scrum.org/PSM

## Scrum Master Choices




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*"A person with a new idea is a crank until the idea succeeds."*

*- Mark Twain*



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# Theory and First Principles

A Day in the Life...

# Meeting the Team

PURPOSE

Explore how variables lead to complexity

5 minutes

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You are the new Scrum Master for a team that tells you about the terrible temperature in their room.

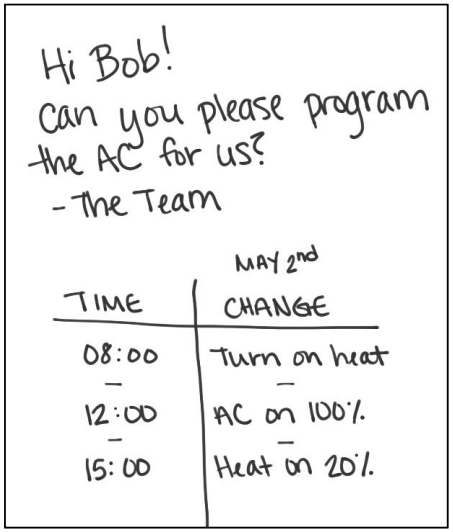
Bob, from the central building services, needs to program the heating, air conditioning, venting, and blinds throughout the day.

You work with the team on assembling a list with all the variables that influence the room temperature to program the climate system upfront. No adjustments are possible during the day.

The team wants a constant and comfortable room temperature.

## Question: What variables will you take into account?

(hint: number of people)




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## There Is a Simple Solution

Variables can be ignored by using an empirical process:

- Transparency is needed to make sure the real temperature is inspected
- Inspect the room temperature at the right frequency
- Adapt the systems that drive the temperature (heating and air cooling) from a common agreement





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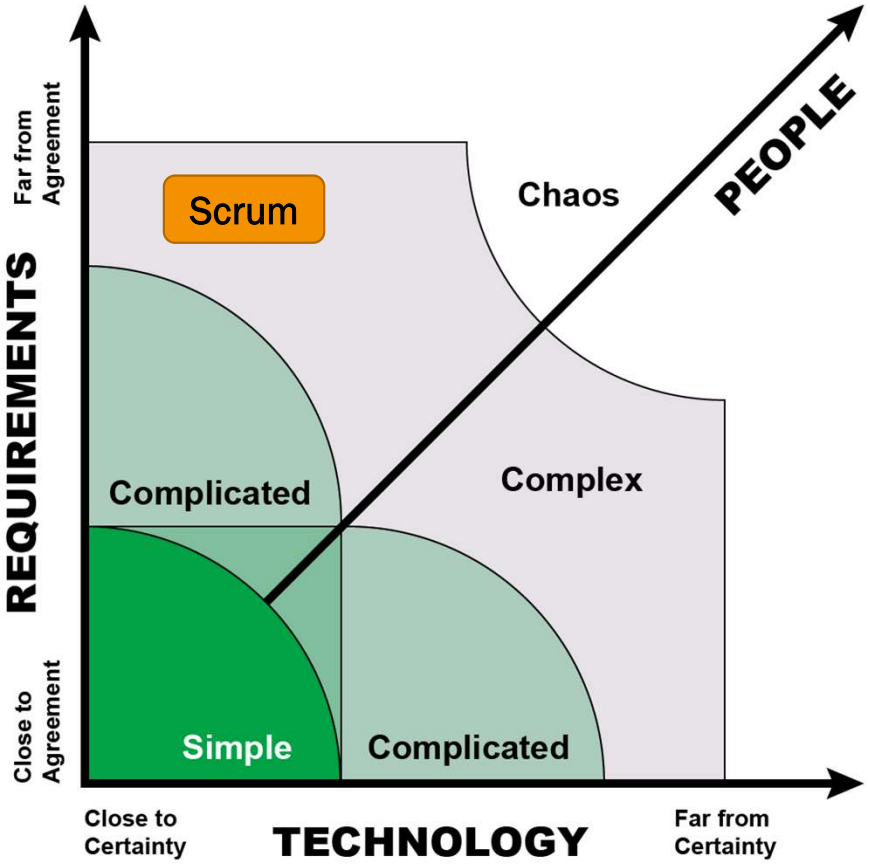

List the variables and parameters that have to be considered in product development.

How predictable are they?

What would you do to control them?

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# The Complexity of Product Development




- Simple
  *everything is known*
- Complicated
  *more is known than unknown*
- Complex
  *more is unknown than known*
- Chaotic
  *very little is known*

Source: Ralph Stacey, University of Hertfordshire


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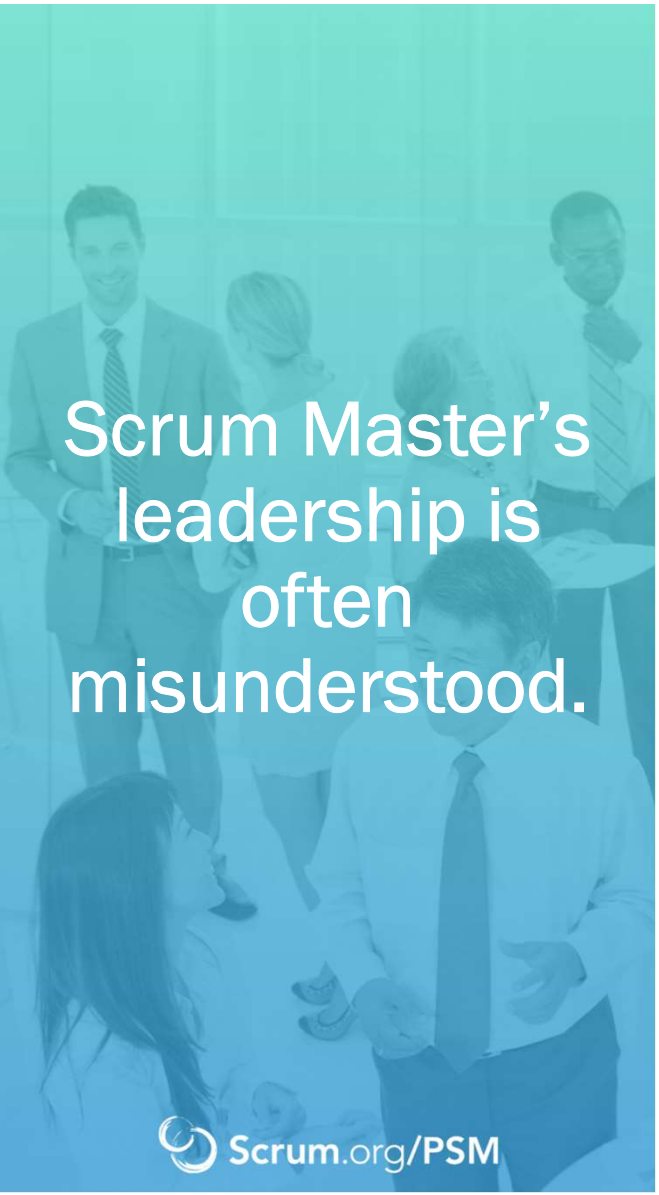
# Relating Complexity to Management Style

| Environment | Characteristics | Leader's Job |
|---|---|---|
| Chaotic | • High turbulence<br>• No clear cause-and-effect<br>• Unknowables<br>• Many decisions and no time | • Immediate action to re-establish order<br>• Prioritize and select actionable work<br>• Look for what works rather than perfection<br>• Act, sense, respond |
| Complex | • More unpredictability than predictability<br>• Emergent answers<br>• Many competing ideas | • Create bounded environments for action<br>• Increase levels of interaction and communication<br>• Servant leadership<br>• Generate ideas<br>• Probe, sense, respond |
| Complicated | • More predictability than unpredictability<br>• Fact-based management<br>• Experts work out wrinkles | • Utilize experts to gain insights<br>• Use metrics to gain control<br>• Sense, analyze, respond<br>• Command and control |
| Simple | • Repeating patterns and consistent events<br>• Clear cause-and-effect<br>• Well established knowns<br>• Fact-based management | • Use best practices<br>• Extensive communication not necessary<br>• Establish patterns and optimize to them<br>• Command and control |



Source: Snowden and Boone. 2007. *A Leader's Framework for Decision Making*

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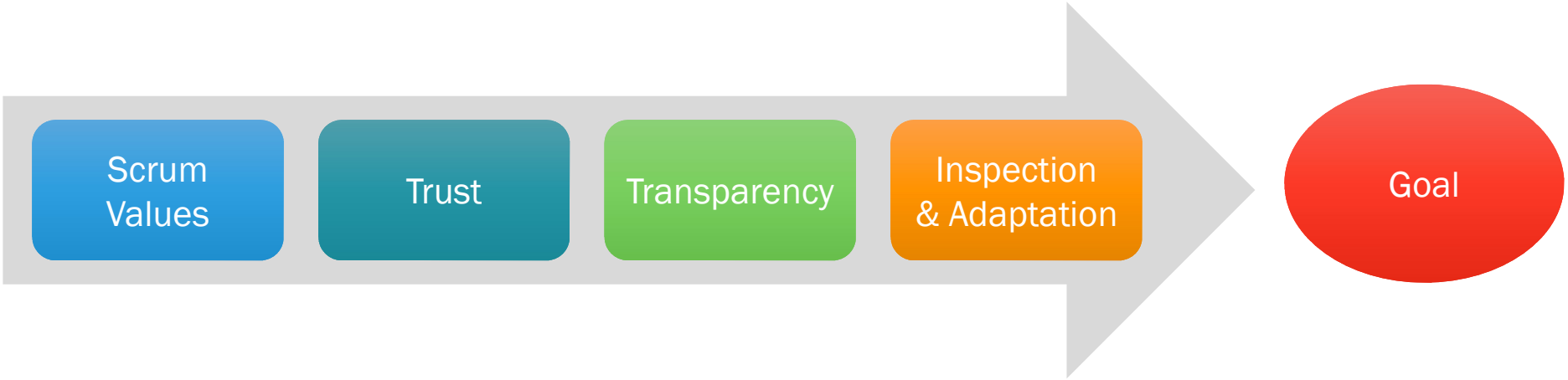



- Success measured by the growth and success of others.
- Influence individuals and teams to take greater responsibility for actions and outcomes.
- Lead without using authority or force; people choose to follow.
- Inspire others to higher greatness.
- Leverage Scrum Values to promote a psychologically safe environment.

Adapted from Robert K. Greenleaf essay The Servant as Leader

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# Empirical Processes Require Trust & Courage





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A Day in the Life...

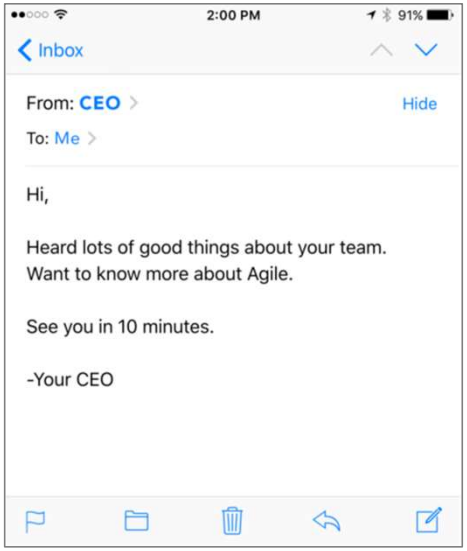
# Meeting the Management

PURPOSE

Exploring the essential advantages of Agility

10 minutes

Scrum.org/PSM

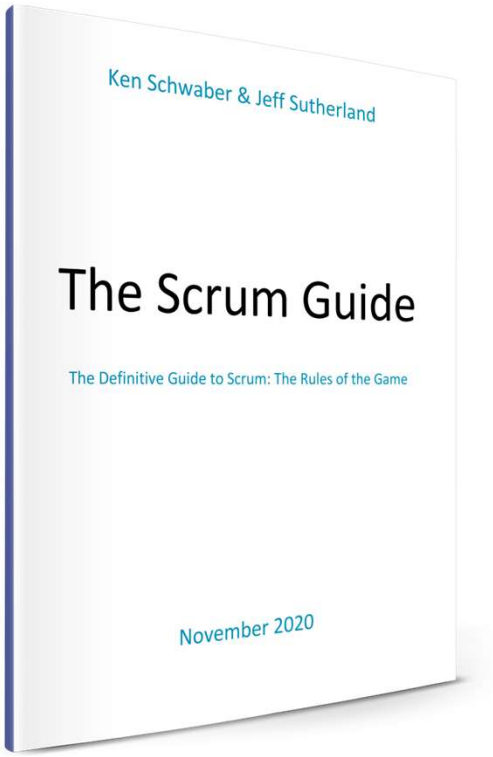





## Explain to the CEO what 'Agile' is about.

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# Definition of Scrum




www.scrumguides.org

Scrum (*noun*):

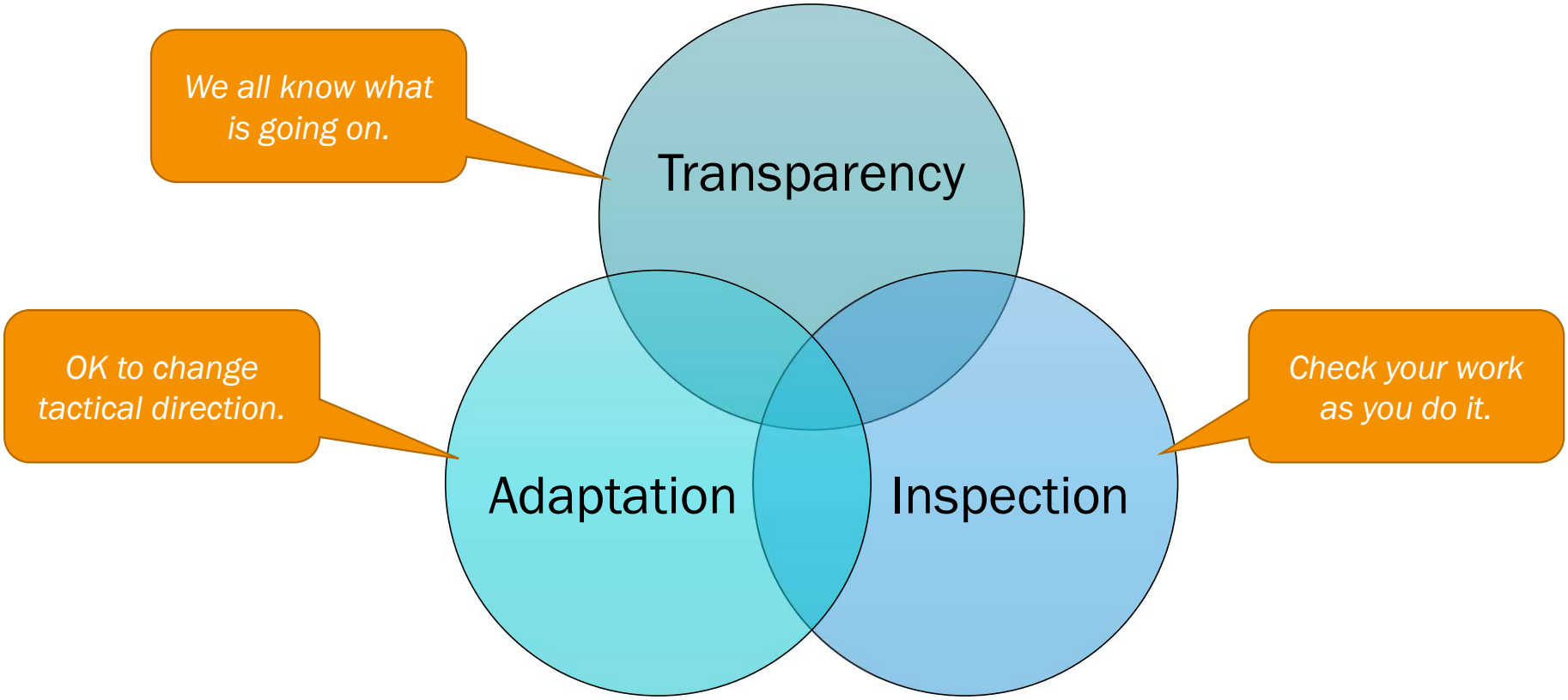
A lightweight framework that helps people, teams and organizations generate value through adaptive solutions for complex problems.

Scrum is

- Lightweight tool for enabling business agility
- Simple and purposefully incomplete


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# Scrum Implements the Three Pillars of Empirical Process Control





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# Scrum Is a Foundation for Agility

Limit risk, provide transparency and be able to adapt through short, high value iterations:

- To deliver valuable, opportunistic pieces of functionality frequently.
- By self-managing, cross-functional teams.



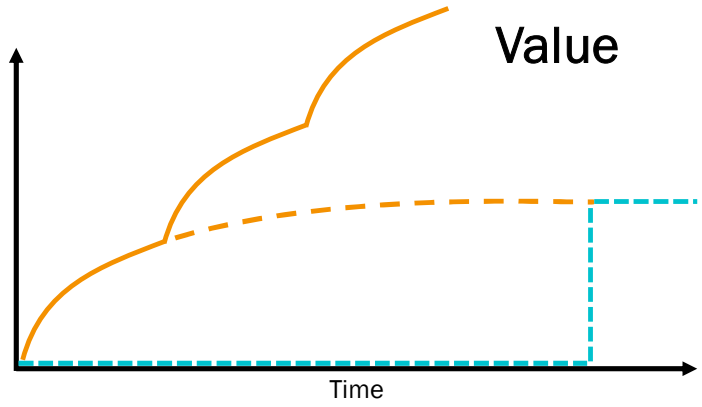


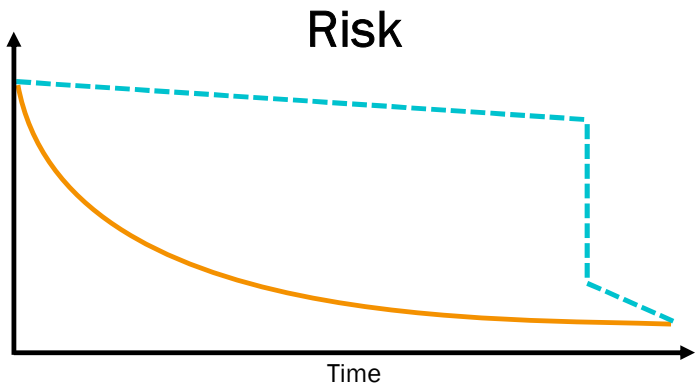


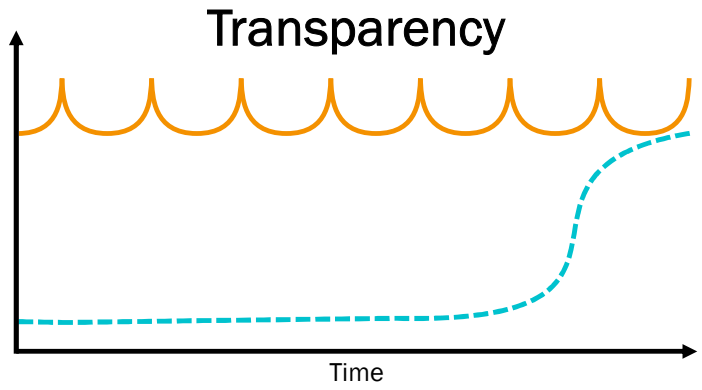

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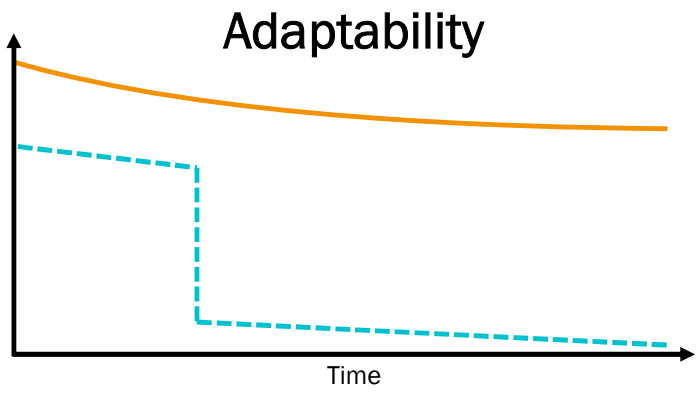
# Comparing Evolutions

Traditional | Scrum





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# Scrum: What's in a Name?





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Exercise

# Is It Customer Service?

PURPOSE

Explore the impact of courage and transparency

10 minutes

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You are a student working your way through college. You work at Burger Kitchen earning minimum wage. You are on the 2pm to 11pm shift, and the only person on duty.

You are cleaning up at 10:30pm when a customer approaches and orders a Double Burger Kitchen Deluxe with onions, cheese, and bacon and an order of fries. The price is $6. The customer informs you that they only have $1.20.

- Burger Kitchen is high quality. Everything is cooked from scratch.
- There is no pre-cooked food you were planning on throwing out.
- Burger Kitchen uses strict inventory control. Anything you take to give to the customer will be charged to your paycheck.
- You have not yet entered the order.

**Question: What do you do? What do you tell the customer?**

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# TAKE AWAY

## Theory and First Principles

- Product development resides in the complex domain
- The best fit for complexity is the empirical process
- The 3 pillars of empiricism are transparency, inspection and adaptation
- Transparency requires trust and courage

Scrum.org/PSM

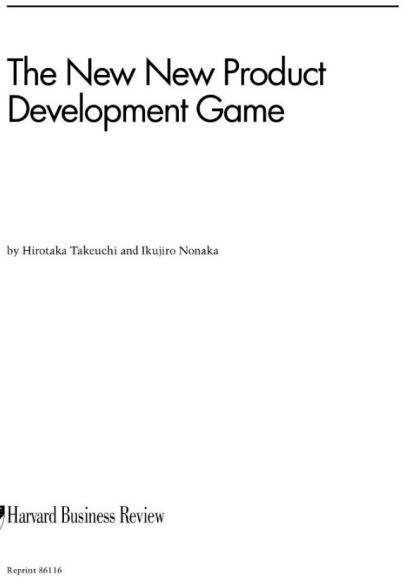
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# Suggested Reading

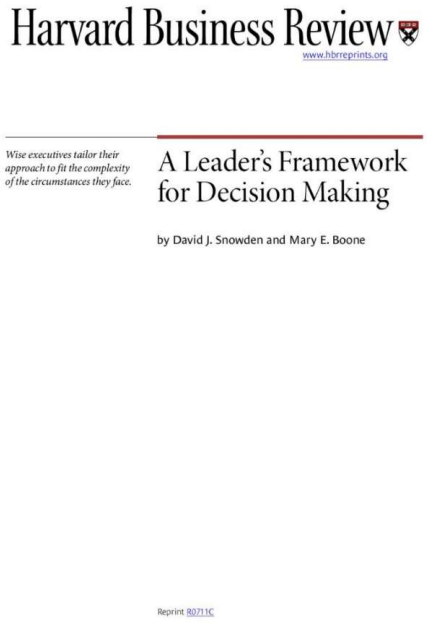
**"The Scrum Guide"** (Ken Schwaber, Jeff Sutherland)




**"The New New Product Development Game"** (Takeuchi, Nonaka)




**"A Leader's Framework for Decision Making"** (Snowden, Boone)





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*"A man who carries a cat by the tail learns something he can learn in no other way."*

*- Mark Twain*

3

# The Scrum Framework




Do you know the elements of the Scrum framework?

| Team | Artifacts | Events |
|---|---|---|
| • | • | • |
| • | • | • |
| • | • | • |
| | | • |
| | | • |

What else do you associate with Scrum?

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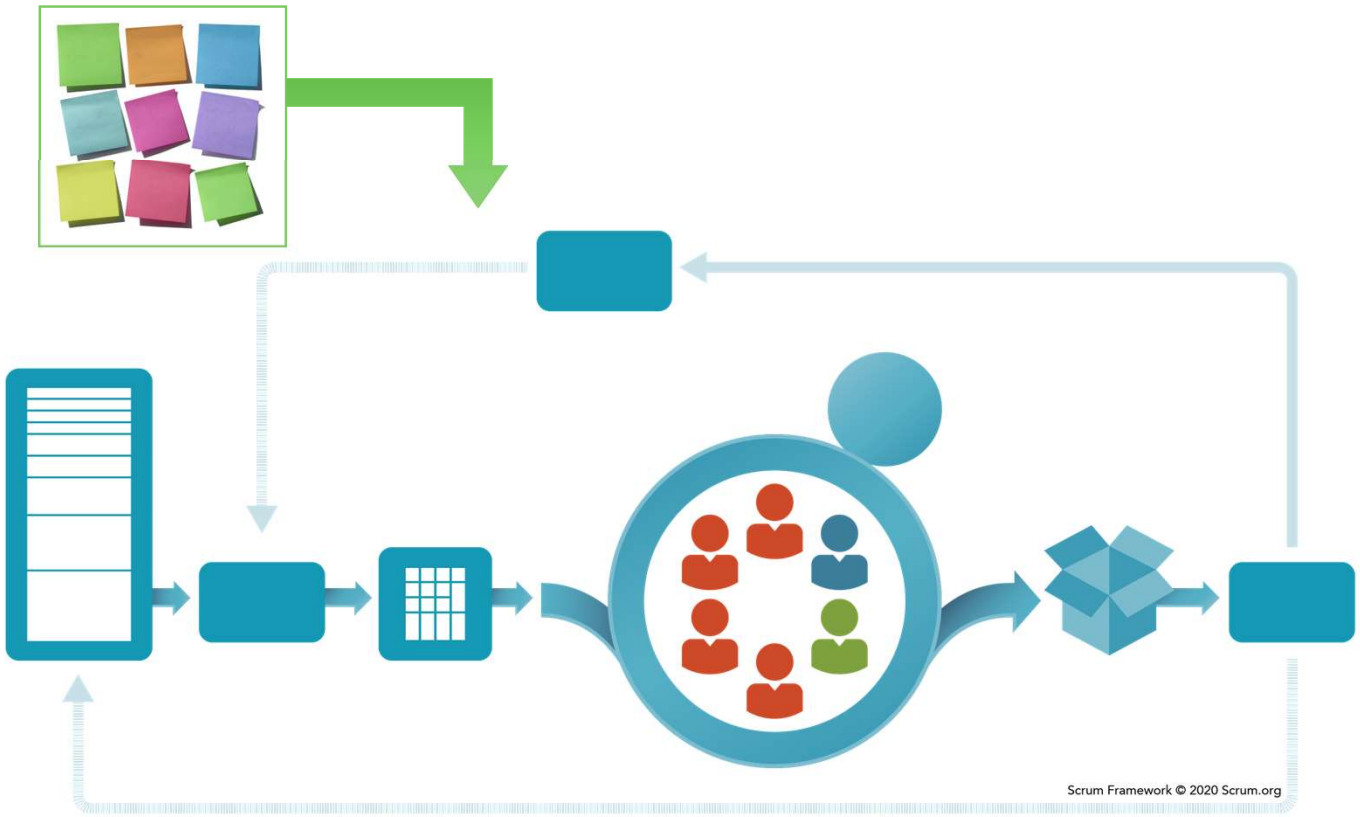
Exercise

# Fitting the Pieces Together

15 minutes

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Each student, add an element of Scrum to the following scheme:




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# Accountabilities, Artifacts, Events, Commitments, and Other Concepts

**Accountabilities**

- Product Owner
- Developers
- Scrum Master

**Artifacts**

- Product Backlog
- Sprint Backlog
- Increment

**Events**

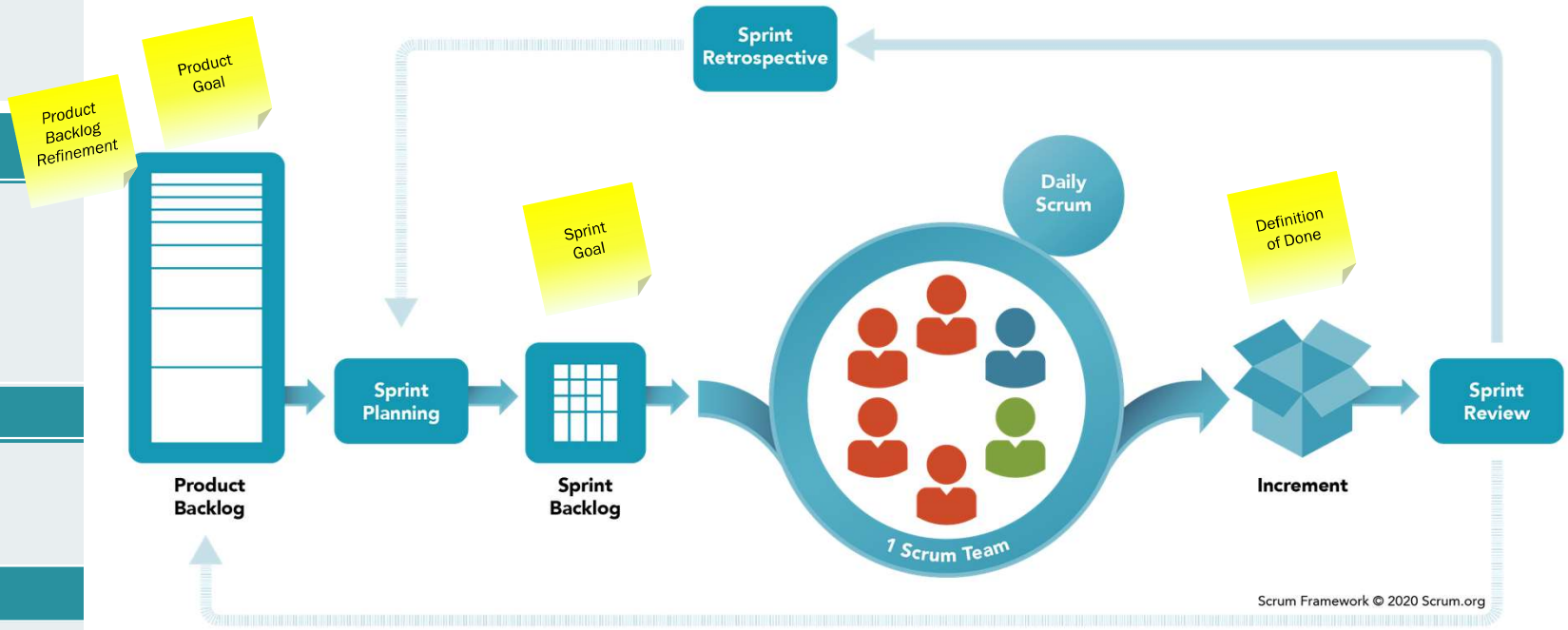
- Sprint
- Sprint Planning
- Daily Scrum
- Sprint Review
- Sprint Retrospective

**Commitments**

- Product Goal
- Sprint Goal
- Definition of Done

***Other***

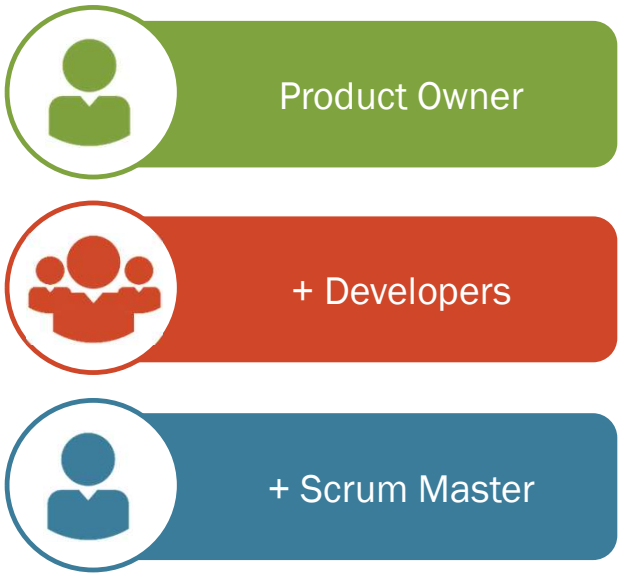
- *Product Backlog Refinement*






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# Scrum Team Includes Specific Accountabilities




- Maximizes value of the product
- Manages the Product Backlog

- Develop quality solutions
- Manage the Sprint Backlog

- Enables Scrum Team effectiveness
- Manages the use of Scrum

- Creates Increments every Sprint
- Is self-managing & cross-functional


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A Day in the Life...

# Judi Is in Trouble

PURPOSE

Demonstrate Scrum accountabilities

5 minutes

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Your friend Judi is in trouble. Judi is CEO of a community portal in San Francisco. The portal has over 20m subscribers, of whom about a million are always active.

The portal has not been updated with new functionality for over 5 months. Only news and data are updated.

There are five Product Managers, all vice presidents, responsible for advertising, dating, community, vacations, and classified functionality. They each receive commissions on the revenue from their respective sections of the portal.

## Question: She asks you for a recommendation to fix this.

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A Day in the Life...

# David Saves the Day

PURPOSE

Demonstrate Scrum accountabilities

5 minutes

Scrum.org/PSM

David is Product Owner at Sprint Planning.

He presents a Product Backlog different from what he and the other Product Managers agreed on.

After more than 3 hours of bickering, David and the Product Managers are nowhere.

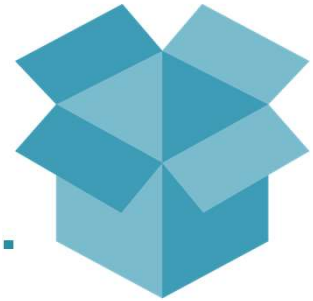
Question: You are there to help them get started. What do you suggest?

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# A Sprint Is an Agreement

| SCRUM TEAM | STAKEHOLDERS |
| --- | --- |
| "Every Sprint you can have us do something new as you see fit." | "We leave you alone to let you work on what we need most." |
| FLEXIBILITY | STABILITY |

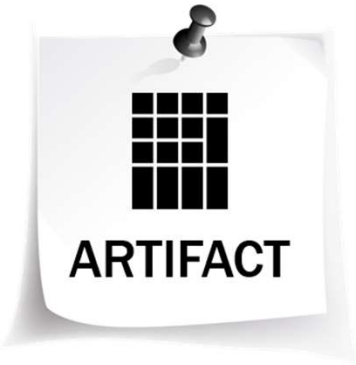
The entire point of Scrum is to create a Done Increment.



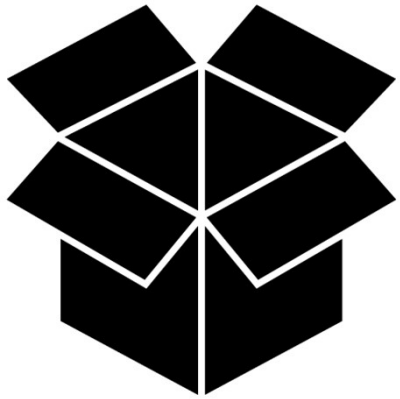


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## Increment




- The working product
- Concrete stepping stone toward the Product Goal
- Must meet the Definition of Done
- Done = Increment must be usable to provide value



**Commitment: Definition of Done**
A description of the state of the Increment when it meets the quality measures


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During Sprint Planning, the Developers don't know how much Product Backlog to forecast.

What should they take into account?

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A Day in the Life...

# What to Select During Sprint Planning?



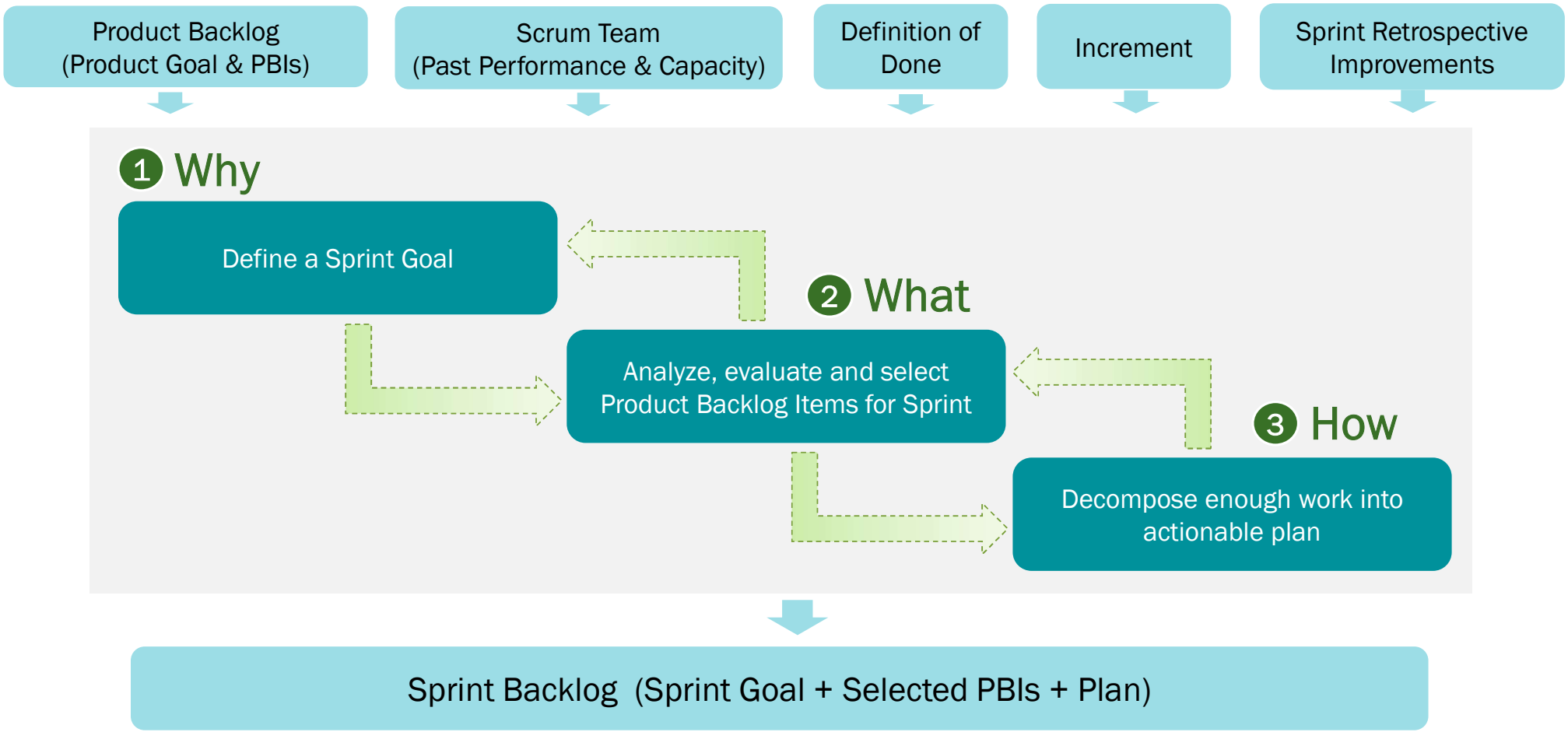

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During the Sprint Planning, the Scrum Team has a hard time figuring out which Product Backlog Items to include in the current Sprint.

What would you advise them as a Scrum Master in this situation? How would you help them get started?

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# Sprint Planning Flow

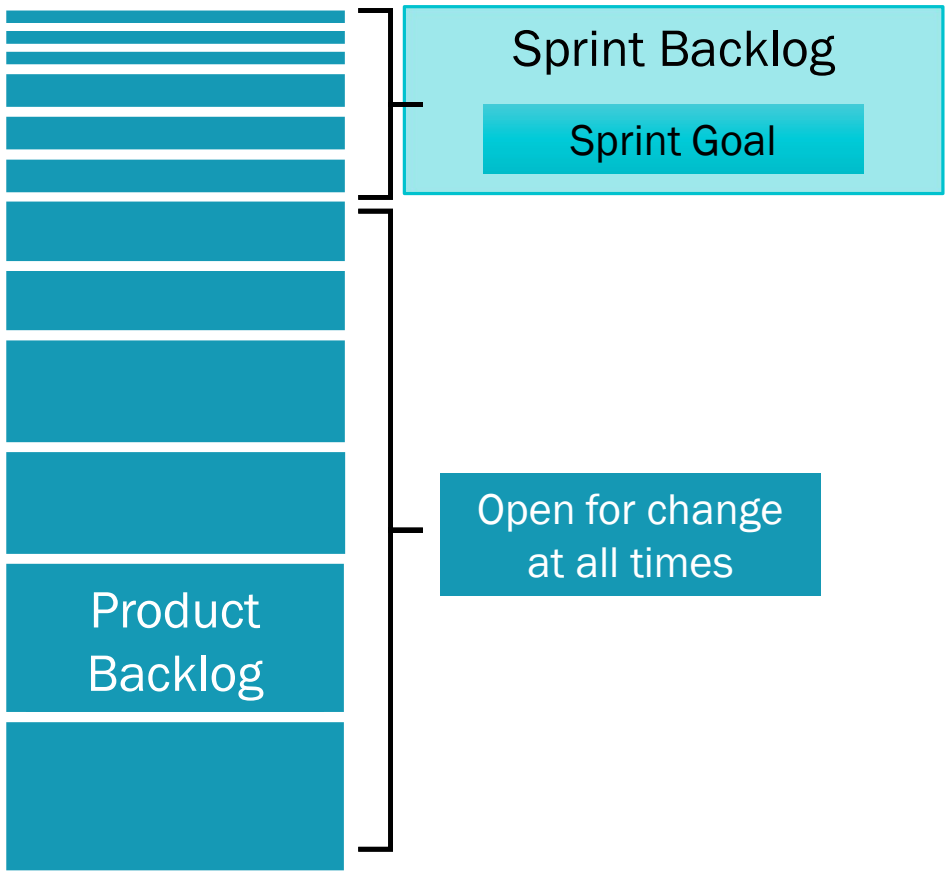




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# Sprint Goal Is a Commitment

- Sprint Goal is the single objective for the Sprint (the **why**).
- It provides guidance, focus and flexibility on how the functionality is implemented.
- Sprint scope may be re-negotiated upon Sprint learning without affecting the Sprint Goal.
- After items are selected to be in the Sprint (the **what**), the remaining Product Backlog will continue to change, evolve, and be refined.

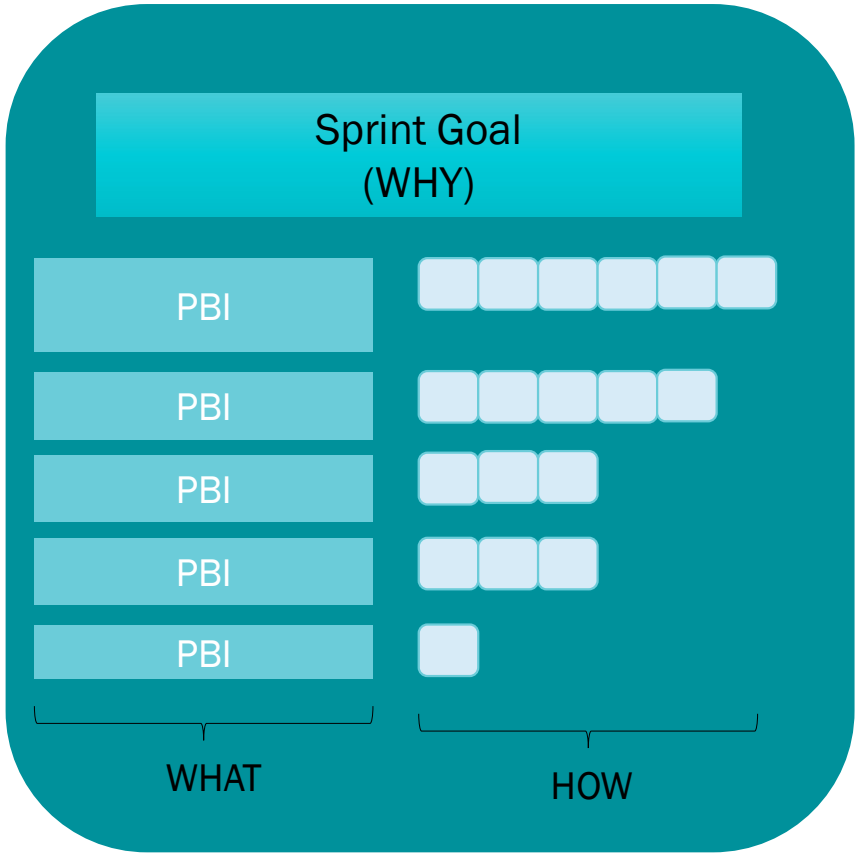




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# Sprint Backlog

- Sprint Backlog consists of the Sprint Goal, selected Product Backlog Items and a plan for delivering the Increment.
- It is a highly visible, real-time picture of the work planned in order to achieve the Sprint Goal.
- Developers create and update the Sprint Backlog as they see fit and as more is learned.
- To monitor Sprint progress Developers often use complementary practices (e.g. burndown chart, cumulative flow diagram, or work item aging chart)

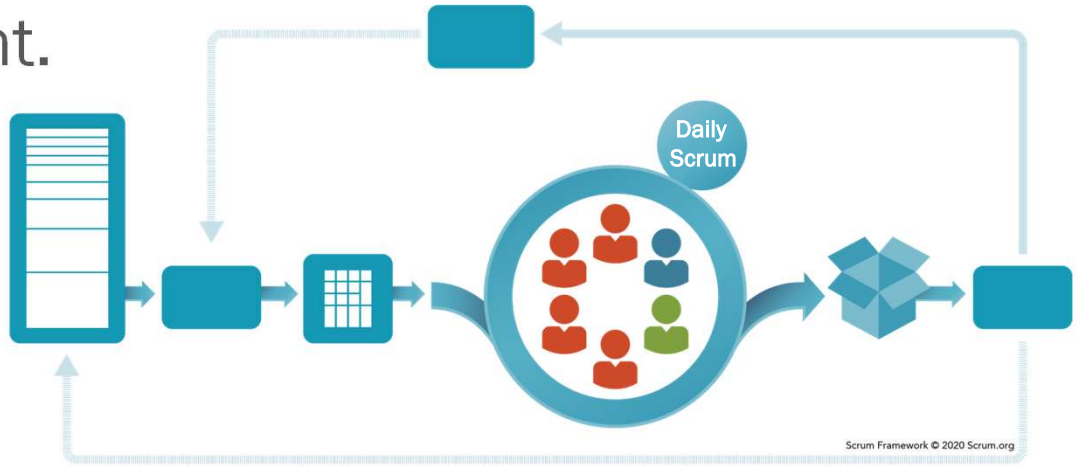




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# Daily Scrum

- 15-minute timebox daily event.
- Consistent place and time.
- Not a problem-solving meeting.
- Not a status meeting.




- Developers inspect their progress toward the Sprint Goal.
- Developers create a plan for the next working day.


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A Day in the Life...

# Burndown as Expected?

5 minutes

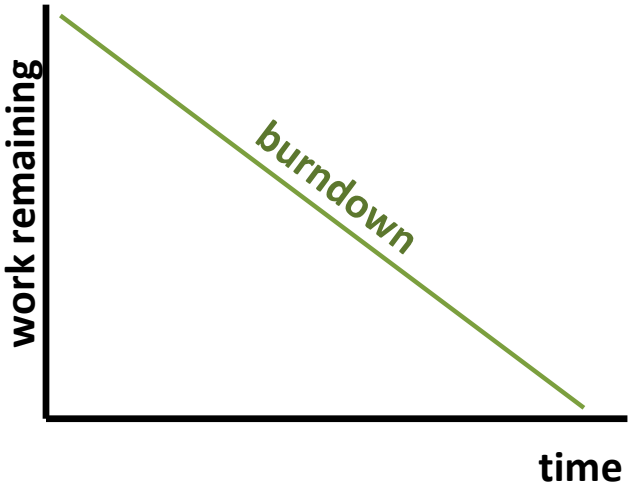
Scrum.org/PSM

The Scrum Master of another team in the company shows you how well his team is doing.

The team is meeting its forecasts and planning well.

He shows the displayed burndown.

## What do you think?




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A Day in the Life...

# A Little More Time

PURPOSE

Examine the value of timeboxing

5 minutes

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Developers are doing well during the Sprint.

However, 3 days before the timebox of the Sprint expires, they request a little more time, 1 or 2 days at most, to get the testing done.

## Question: Do you extend the Sprint?

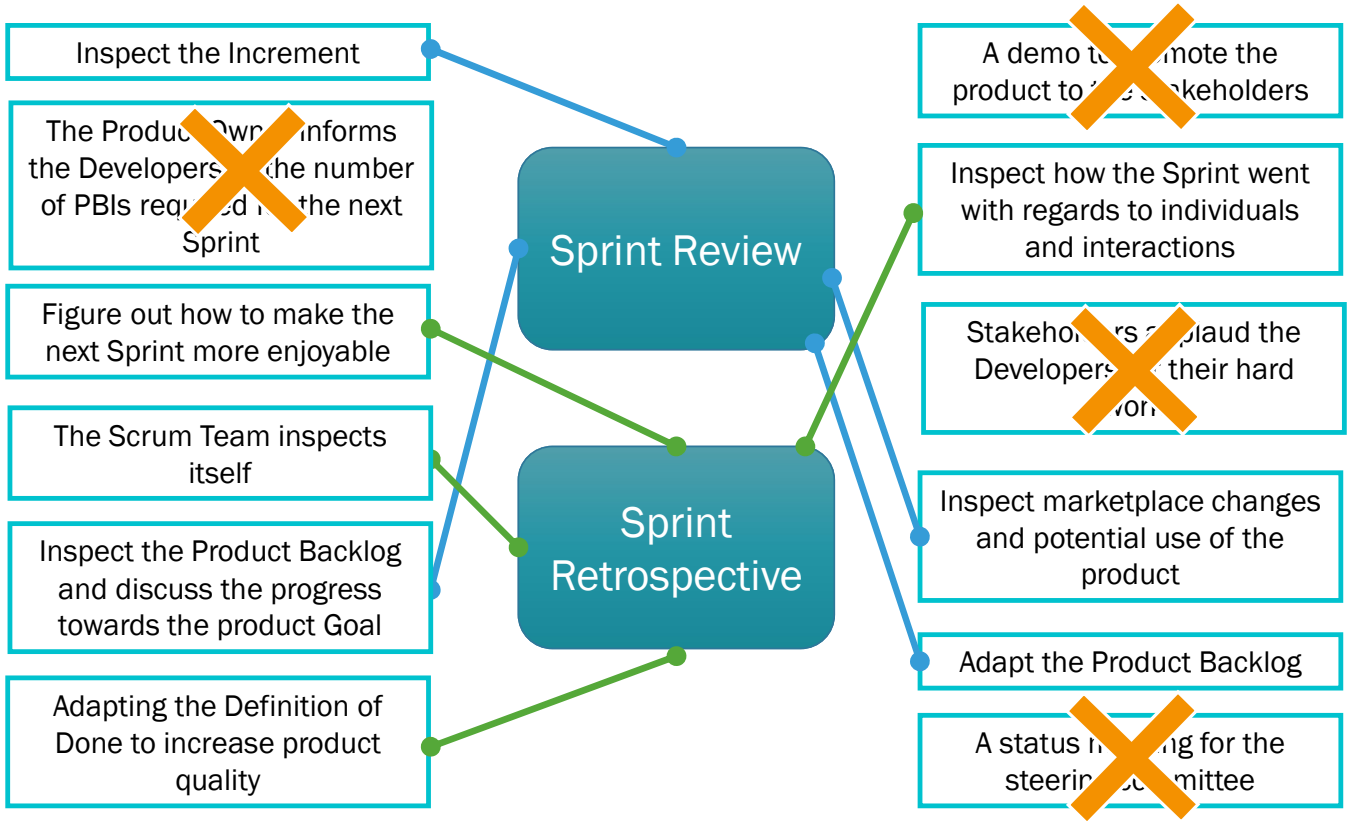
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Exercise

# A Sprint Is a Feedback Loop

5 minutes

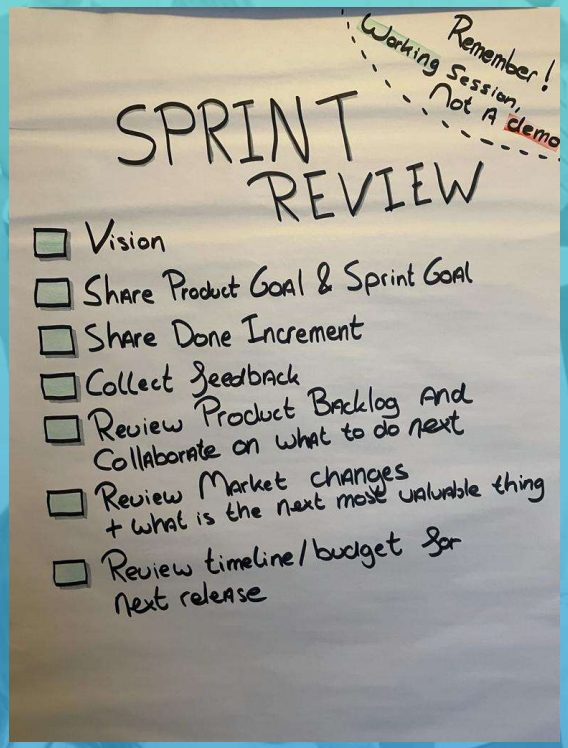
- Connect the statements to the Scrum events.
- Identify common misconceptions.




Scrum.org/PSM

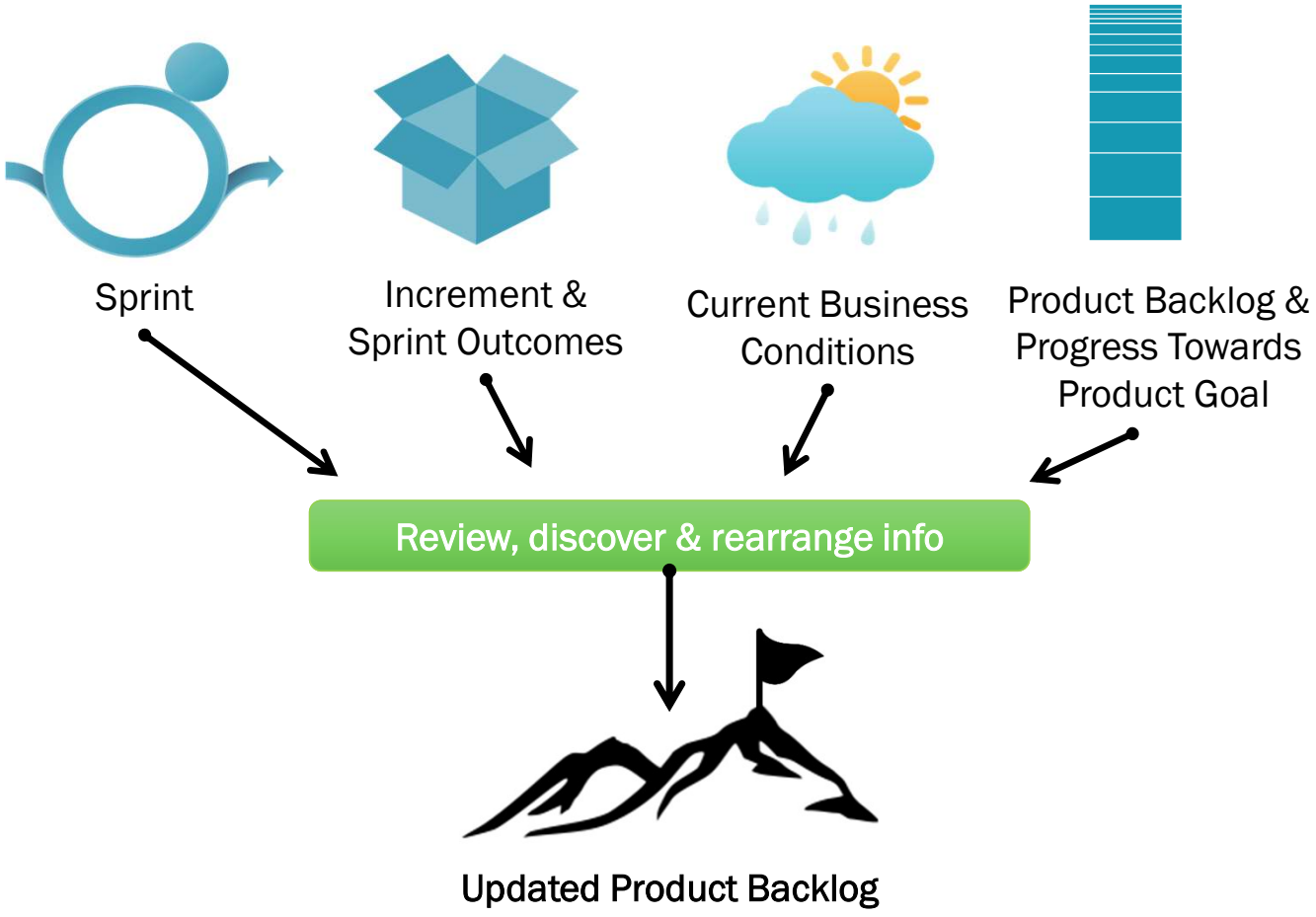
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This is a collaborative working session, not a demonstration.



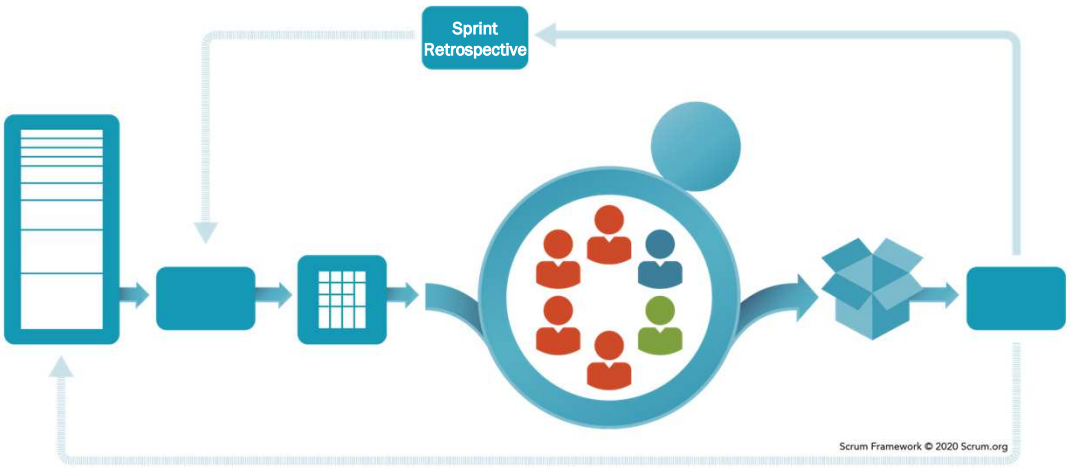

Scrum.org/PSM

# Flow of the Sprint Review




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# Sprint Retrospective




- Scrum Team inspects how the last Sprint went.
  - Individuals, interactions, process, tools
  - Definition of Done
- Scrum Team adapts by identifying the most helpful changes to improve its effectiveness.


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# Scrum Events Quick Reference

| Event | Inspection | Adaptation | Who Participates | Timebox for 1 Month |
|---|---|---|---|---|
| **Sprint Planning** | Product Backlog, Product Goal, Definition of Done | Sprint Backlog, Sprint Goal | Scrum Team | 8 hours |
| **Daily Scrum** | Progress toward Sprint Goal | Sprint Backlog | Developers | 15 minutes (always) |
| **Sprint Review** | Increment, Sprint, Product Backlog, Progress toward Product Goal | Product Backlog | Scrum Team Stakeholders | 4 hours |
| **Sprint Retrospective** | Sprint, Definition of Done | Actionable improvements, Definition of Done | Scrum Team | 3 hours |

Every element of Scrum serves empiricism.


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A Day in the Life...

# Call from the CEO

PURPOSE: Understand how to assess the success of Scrum

5 minutes

Scrum.org/PSM

After a couple of Sprints, the CEO calls you and asks whether Scrum is working.



Question: How do you answer this question? What elements are you going to consider?

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# TAKE AWAY

## The Scrum Framework

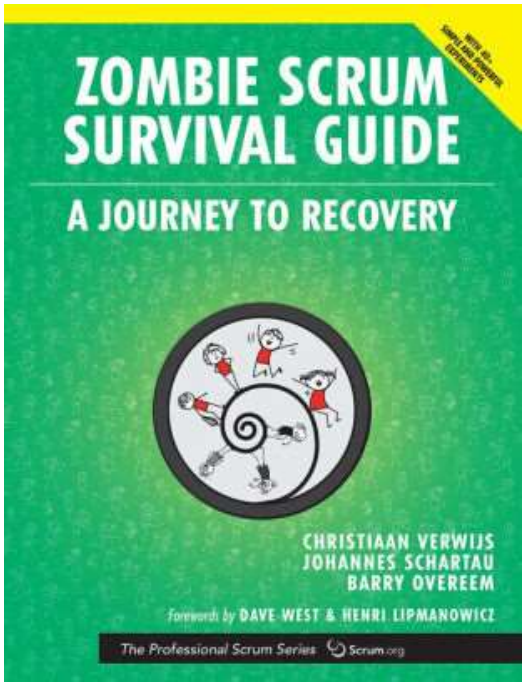
- Scrum implements empiricism in complex environments.
- Scrum Team includes specific accountabilities.
- The Scrum artifacts provide transparent information.
- The Scrum events serve transparency, inspection and adaptation.

Scrum.org/PSM

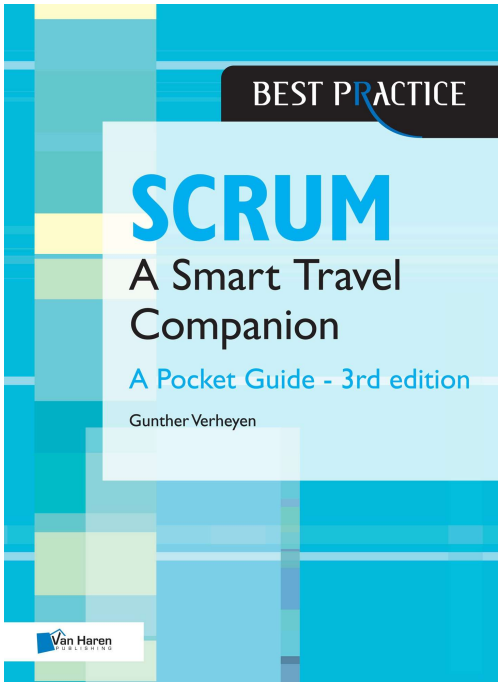
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# Suggested Reading

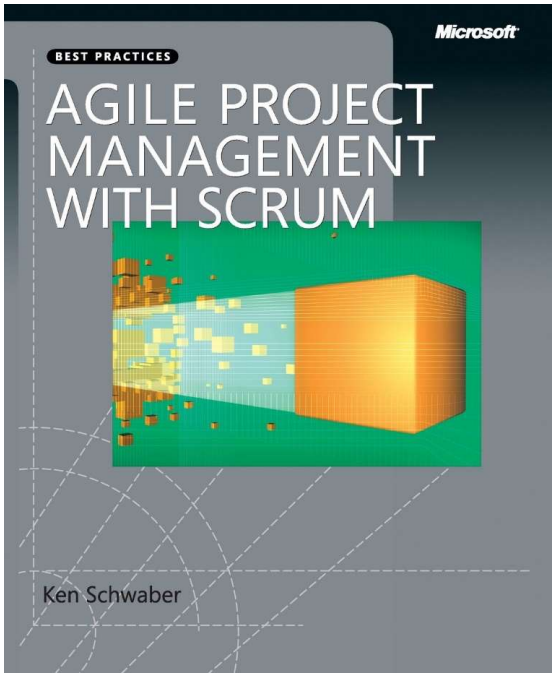
"Zombie Scrum Survival Guide" (Christiaan Verwijs, Johannes Schartau, Barry Overeem)




"Scrum – A Pocket Guide" (Gunther Verheyen)

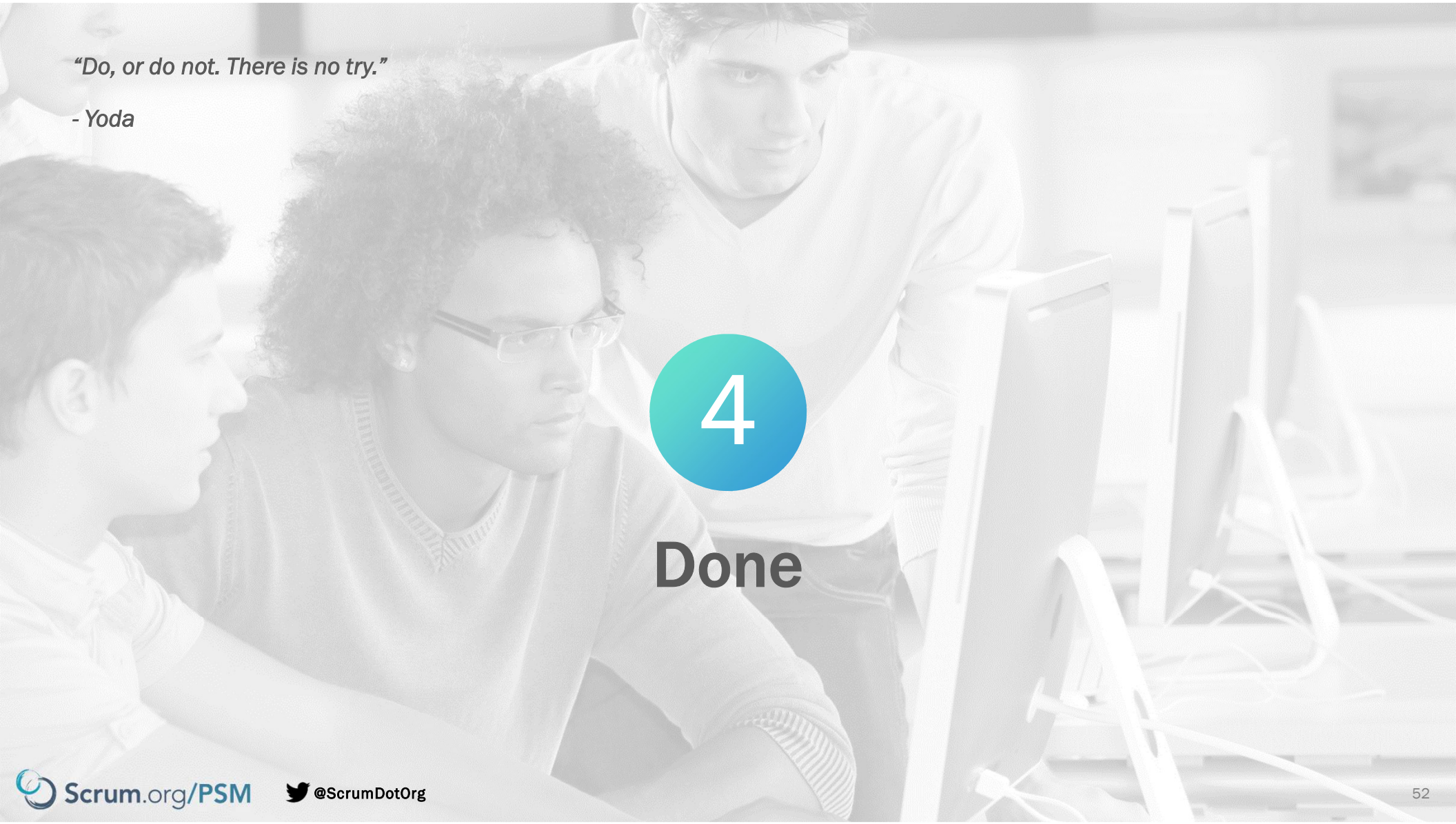



"Agile Project Management with Scrum" (Ken Schwaber)






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*"Do, or do not. There is no try."*

*- Yoda*

4

# Done

Exercise

# Quality, a Habit?

PURPOSE

How the Definition of Done serves transparency

5 minutes

Scrum.org/PSM

Christine is Product Owner. Based on the average work delivered of the previous release (13 PBIs/ Sprint), Christine estimated a new release of the product to take 7 Sprints. Development is 3 Sprints underway. Product Backlog has been stable.

Over these first Sprints, Developers reported completing an average of 9 PBIs/ Sprint, although not all functionality was fully tested. The Developers estimate that the missing testing would have required 10% more time.

Christine considers the current functionality cohesive enough for her users and wants to release it.

## Question: What is the most effective way to proceed?

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## Technical Debt Affects Transparency

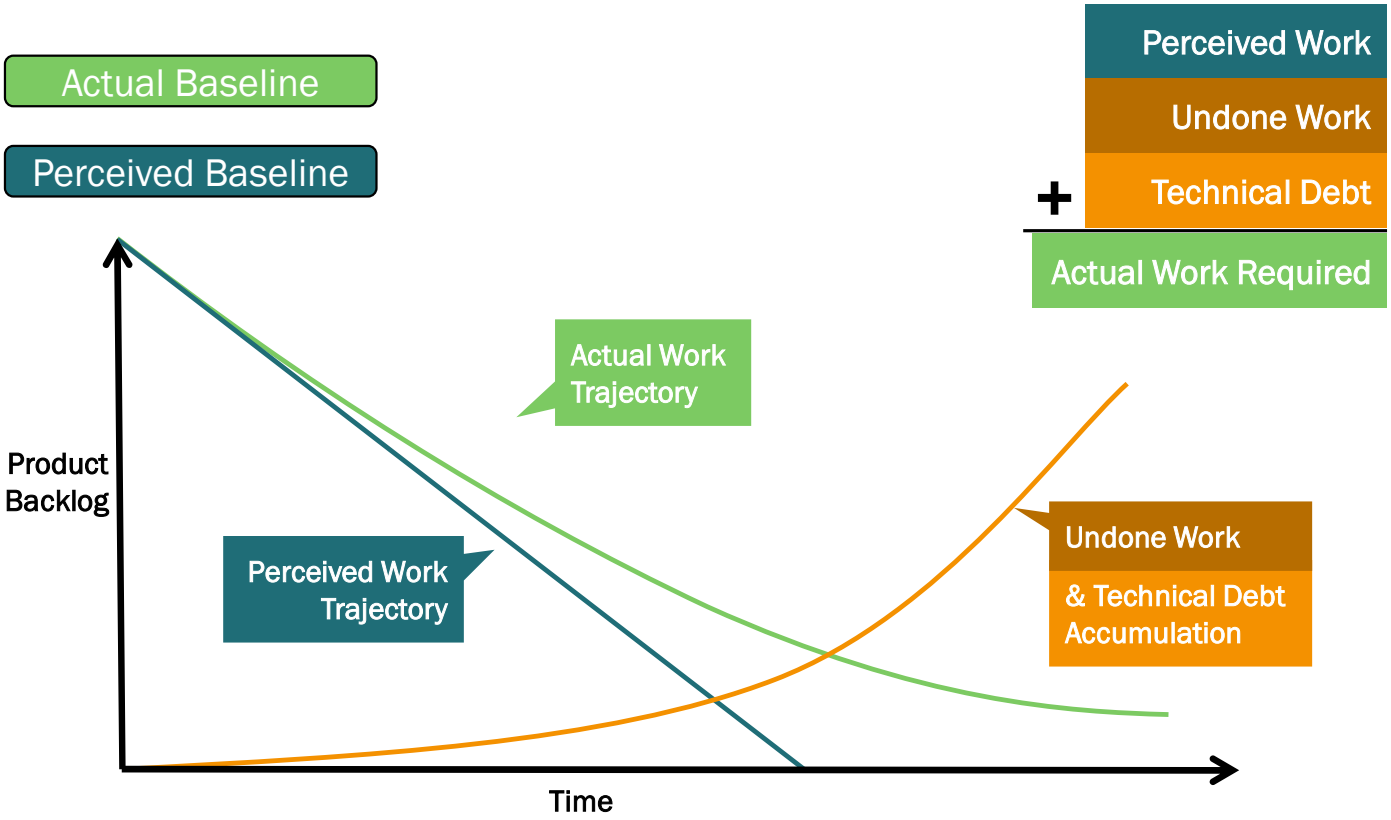
- Technical debt is often the result of decisions made by the Scrum Team to trade quality for speed.
- Technical debt is not undone work.
- It can take many forms.
- It can be incurred consciously or not.

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Remember: Undone Work & Technical Debt do not accumulate linearly

Scrum.org/PSM

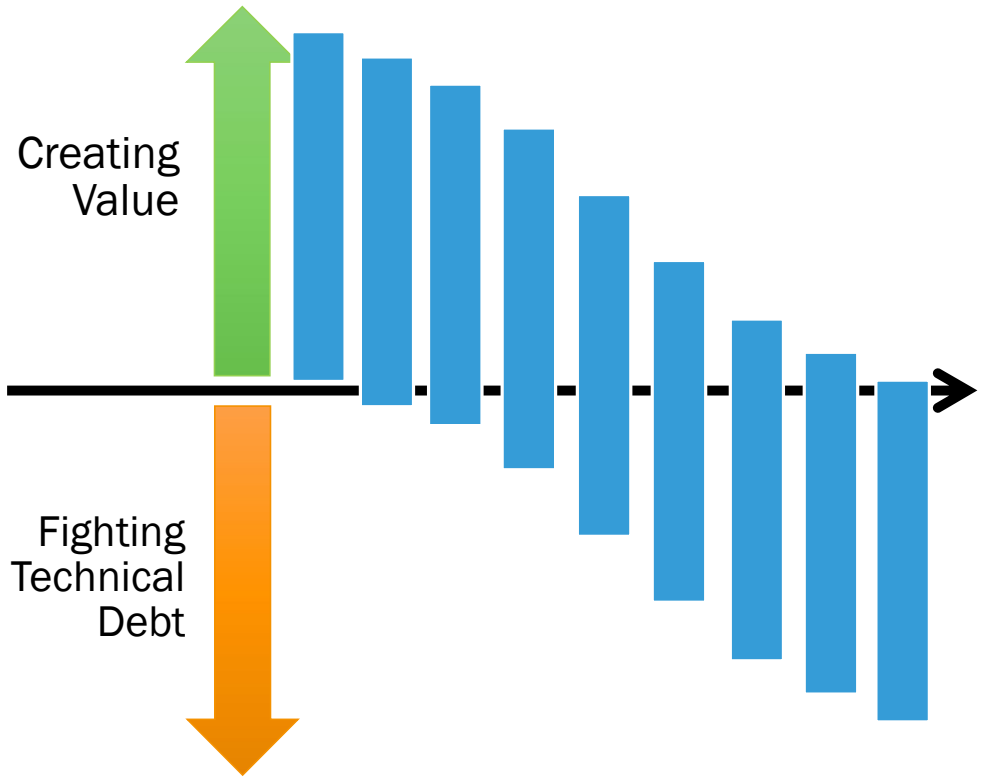
## Undone Work & Technical Debt Increase the Actual Work




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# Dealing with the Technical Debt

- As the technical debt builds up, the ability to create value drops.
- Create transparency about the technical debt in the Product Backlog.
- Improve your Definition of Done to lower the chance for technical debt to be created in the future.
- Stop creating new technical debt and start paying back the existing one, a bite at a time.





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Exercise

# Exploring Definition of Done

5 minutes

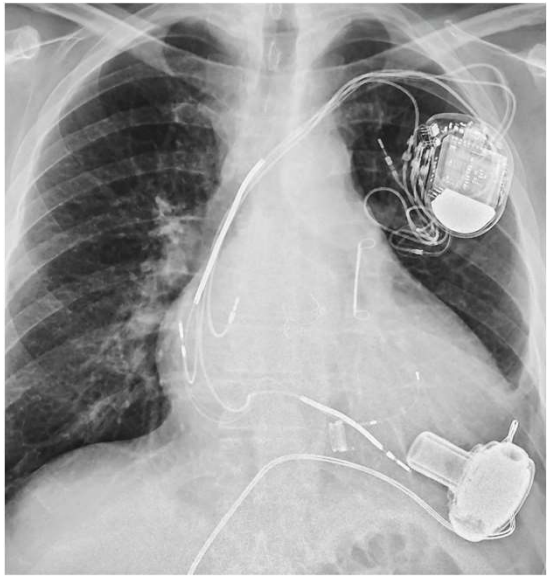
Scrum.org/PSM

Your Scrum Team is one of 7 teams working on a new release of a life-critical product that is shipped internationally.

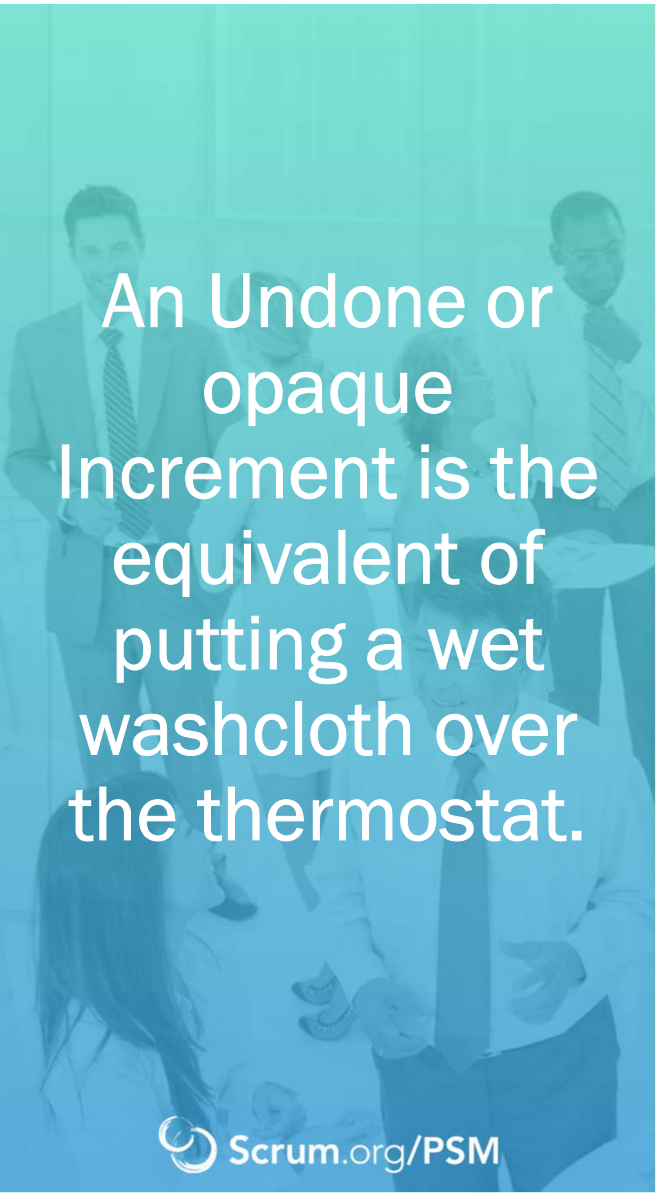
You use 2-week Sprints. Each team has all the skills to fully develop the requirements into a Done Increment.

## Question:

## What would your Definition of Done be? What's so important about it?



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## Did Your Definition of Done Include These?

- Integration testing
- Performance testing
- Stability testing
- Refactoring
- Release notes
- Localizations for the countries where the product will be sold
- Acceptance testing
- User documentation
- Regression testing
- Code reviews

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# Conventions, Standards and Guidelines Serving Done

## PRODUCT QUALITIES

- Quality code base (clean, readable, naming conventions)
- Valuable functionality only
- Architectural conventions respected
- According to design/style guide
- According to usability standards
- Documented
- Service levels guaranteed (uptime, performance, response time)

## DEVELOPMENT STANDARDS

- Pair programming
- TDD, ATDD, BDD
- Refactoring
- UI testing
- Functional testing
- Performance testing
- Continuous Integration/Continuous Deployment

For more on this topic






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Consider your current team at work.

Is your team able to deliver a Done Increment by the end of the Sprint?

If not, how do you get there?

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# TAKE AWAY

**Done**

- If Scrum was to be reduced to one purpose only, it would be the creation of **Done** Increments.
- Done Increments are essential for Scrum's empiricism and agility.
- A robust Definition of Done is a commitment to the Increment's quality and transparency.
- Definition of Done reflects usable.

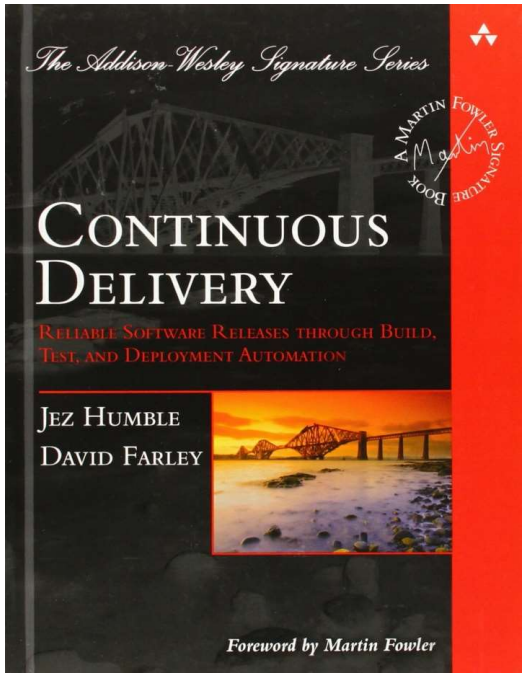




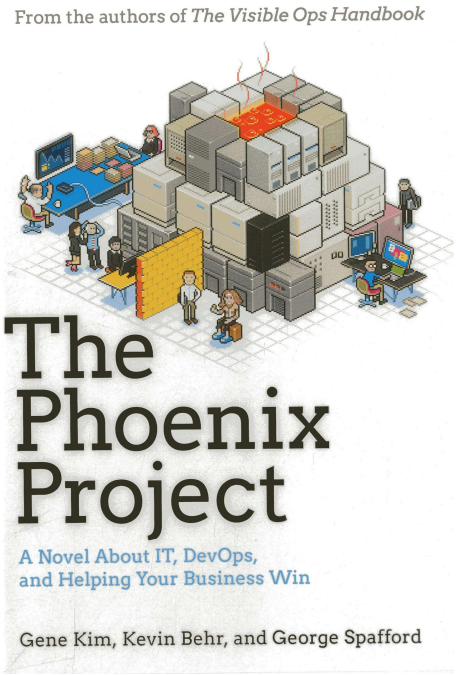

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# Suggested Reading

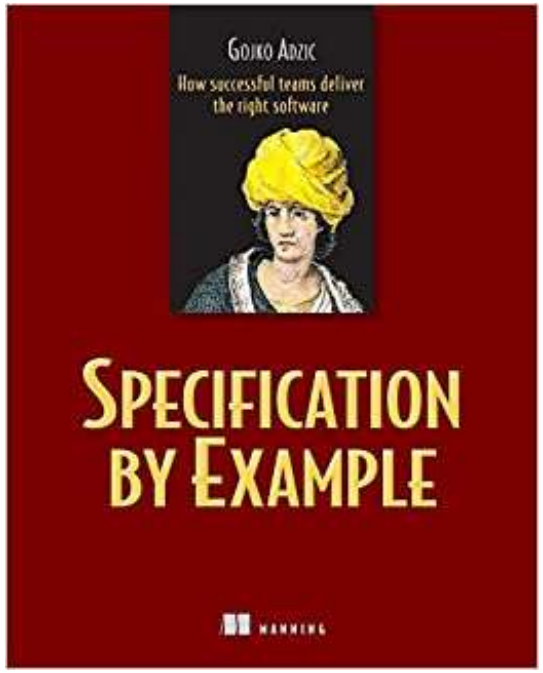
**"Continuous Delivery"**
(Jez Humble, David Farley)




**"The Phoenix Project"**
(Gene Kim, Kevin Behr and George Spafford)




**"Specification by Example"**
(Gojko Adzic)





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*"In life, as in football, you won't go far unless you know where the goalposts are."*

*- Arnold H. Glasgow*

5

# Product Delivery with Scrum

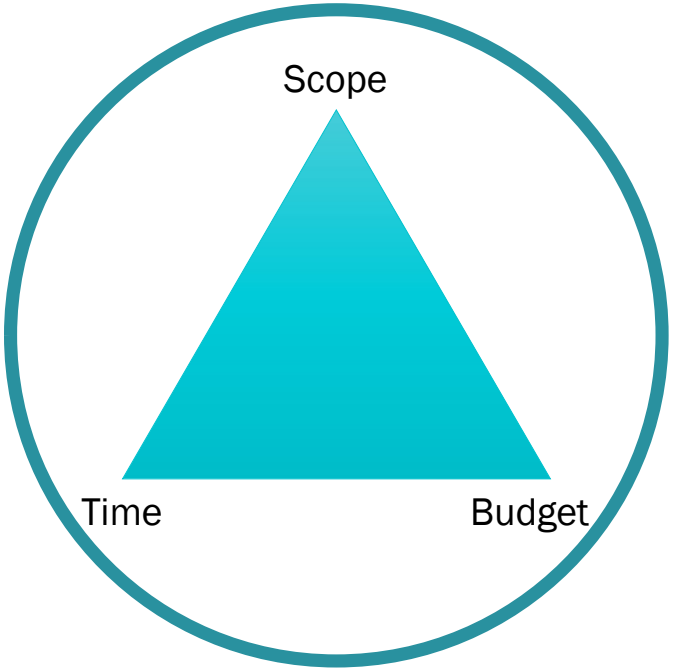
# Product Delivery Requires a Distinct Mindset

## Product Mindset

Success *continuously* driven by *business metrics* outside in:

- User adoption/retention
- Revenue
- Cost savings per feature

Leads to less waste, more creativity, and more releases.




## Project Mindset

Success *upfront* defined inside out:

- Scope
- Time
- Budget

Leads to less business involvement, more task management.

On time, within scope, and within budget doesn't automatically equal product success


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OmniDrive Storyline Information

# OmniDrive Opportunity




It is now November.

There is a market opportunity to provide an extension (retro-fitting) to allow all cars to be automatically driven.

OMNIDRIVE

OmniDrive has secured USD 100 million in venture capital backing, to be released in tranches when key viability milestones are achieved.

The board is looking for an indication of the duration and cost of completing the development.

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**For** all car owners

**Who** want to have an autopilot in their cars,

**The** OmniDrive car guidance solution

**Is an** extension to existing cars

**That** adds auto drive, collision avoidance and adaptive speed control.

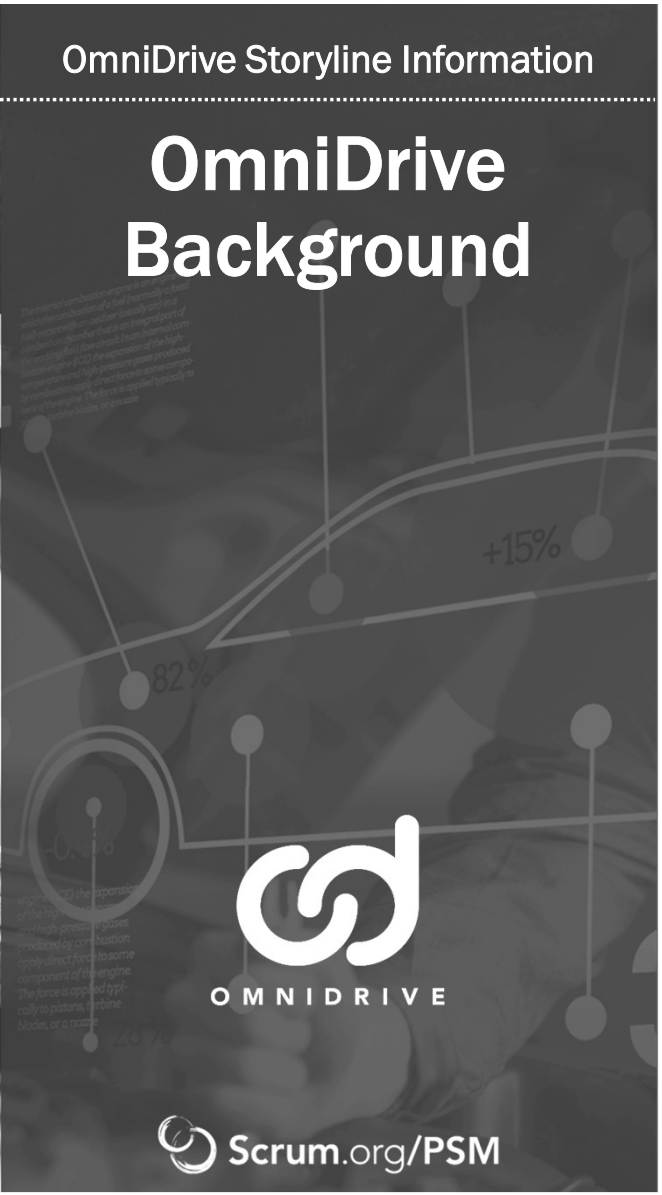
**Unlike** manual driving or the Google car,

**Our Product** does not require buying a new car; you can enhance your current car.

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OmniDrive Storyline Information

# OmniDrive Background




At a press conference on January 15, OmniDrive will announce the following release schedule and release objectives. The venture capital will be released in tranches of USD 20 million, based upon successful completion of the following proof points.

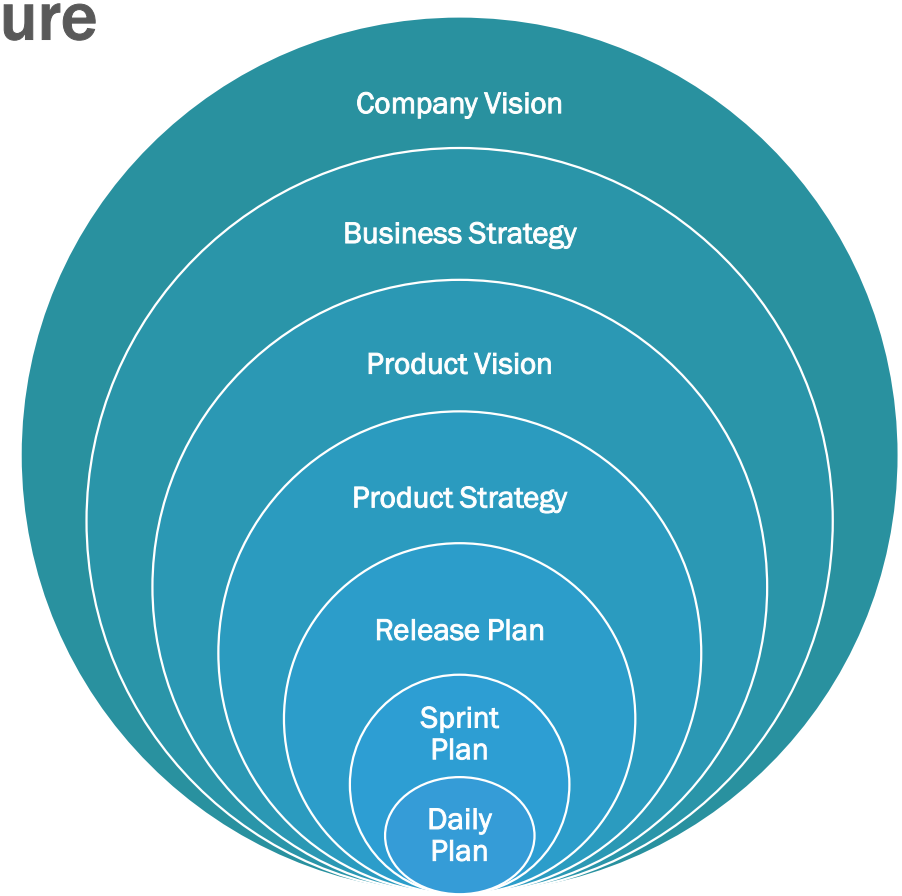
- R1 – March 31 – Working prototype
- R2 – Sept 30 – Driving Assist proven and approved in at least one country
- R3 – Limited Self Drive proven and approved in at least one country
- R4 – Auto Drive proven and market ready

Revenue will be earned by selling market feasible products beginning with R2.

OmniDrive needs to know the likelihood that the working platform will be available by the above dates prior to this press conference.

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# The Bigger Picture




Why do we need goals in Scrum?

Where would you place the Product Goal and the Sprint Goal ?


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The hardware prototype is already available, and your team will have access to the mechanical, electrical, and design engineers who created it.

*(See Case Study Handout)*

## Create a Product Backlog for Release 1:

- Define a Product Goal.
- Create a card for each Product Backlog item.
- Review both functional and non-functional items.
- Prepare to present your Product Backlog to the class.

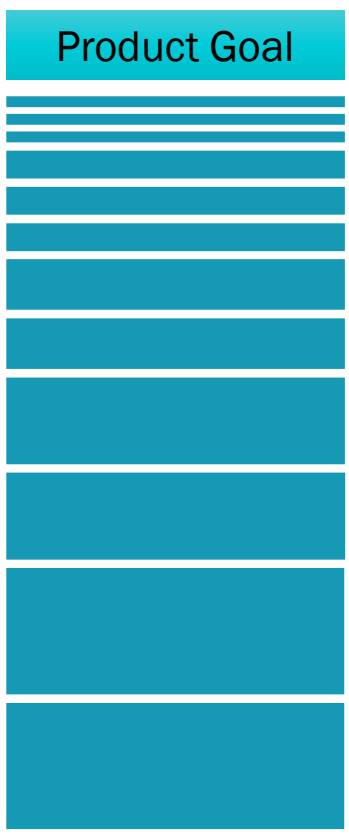
*Do not strive for perfection, just the best you can do!*

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# Product Goal is a Commitment

- Describes a future state of the product that can serve as a target for the Scrum Team to plan against.
- Helps drive the emergence of the Product Backlog.
- Provides focus for longer-term goals and transparency to progress beyond a Sprint.
- The Scrum Team must fulfill (or abandon) one objective before taking on the next.

Product Backlog
Is guided by the
Product Goal but not
limited by it.




For more on this topic



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# Product Backlog Holds the Plan for Future Sprints

- Contains a Product Goal for greater focus and transparency.
- List of what is needed to improve the product.
- Ordered based on:
  - ROI, value, dependencies, risk, and other factors
- Transparent.
- Minimal but sufficient.
- Managed by Product Owner.
- The single source of work for the Scrum Team.





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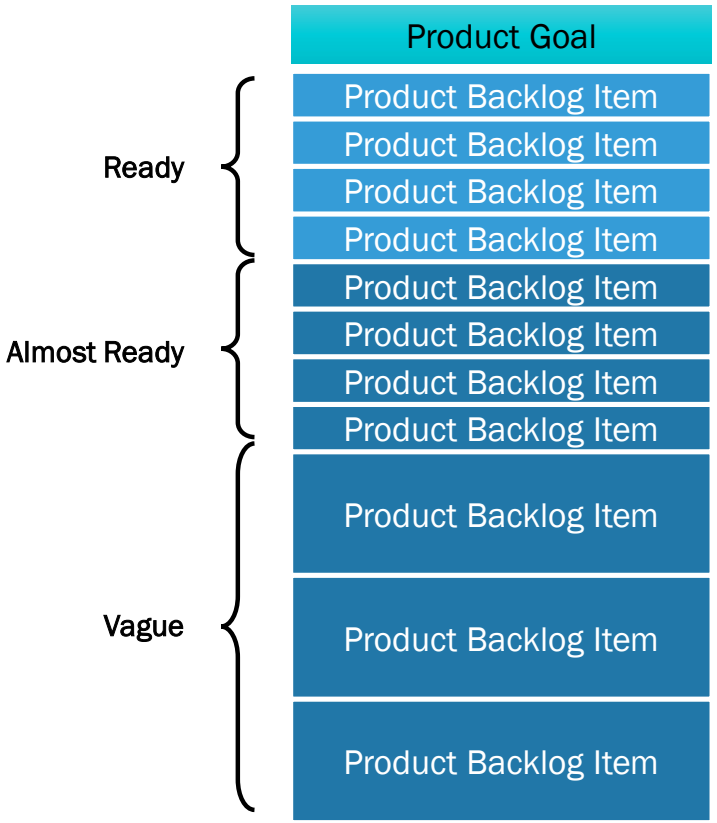

Take a look at your Product Goal and Product Backlog.

- What would be a good candidate for the first Sprint Goal ?
- Which Product Backlog Items should be refined ?

Refine your Product Backlog and be ready to present it to the class.

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# Refining the Product Backlog

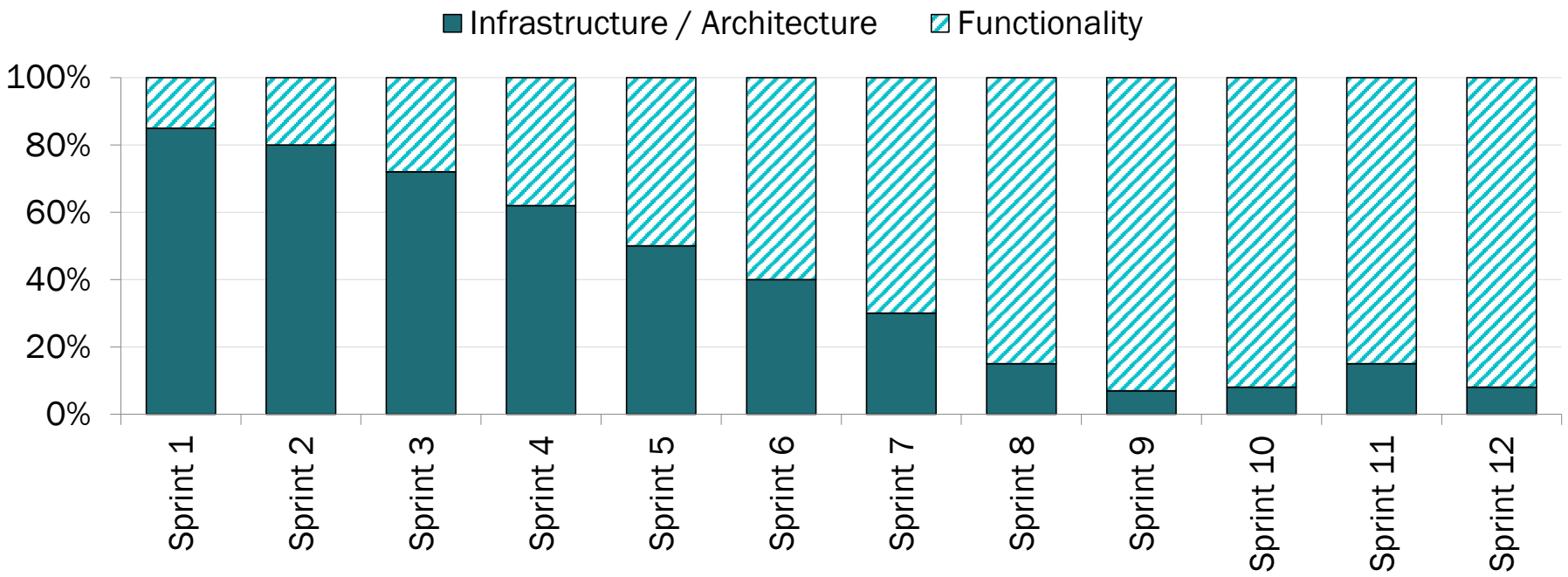



- Product Backlog is continuously refined to increase understanding, granularity and transparency.
- Larger Product Backlog Items are broken down into consumable pieces of value.
- Product Backlog Items that can be Done by the Scrum Team within one Sprint are deemed ready for selection in a Sprint Planning event.


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# Product Backlog Supports Emergent Architecture

Every Sprint must deliver some business functionality.




For more on this topic



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# Techniques for Product Backlog Ordering & Value




500 Value Points

*Allocate points from a fixed total*

Planning Poker

*Assign relative value points (instead of size)*

Buy a Feature

*Innovation Game using money*

20/20 Vision

*Innovation Game for simple ordering*

Thirty Five

*Collaboration activity for ordering*

What other techniques have you seen Product Owners use?

For more on this topic






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# Methods of Product Backlog Organization




Priority

*Either calculated or relative*

Development Cohesion

*Both product and system*

Business Cohesion

*Smaller area of business affected*

Implementation Cohesion

*A work flow, for instance*

Intentions

*Release grouping*

Cohesion simplifies development and implementation



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Exercise

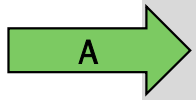
# When Will Item "A" Likely Be Released?

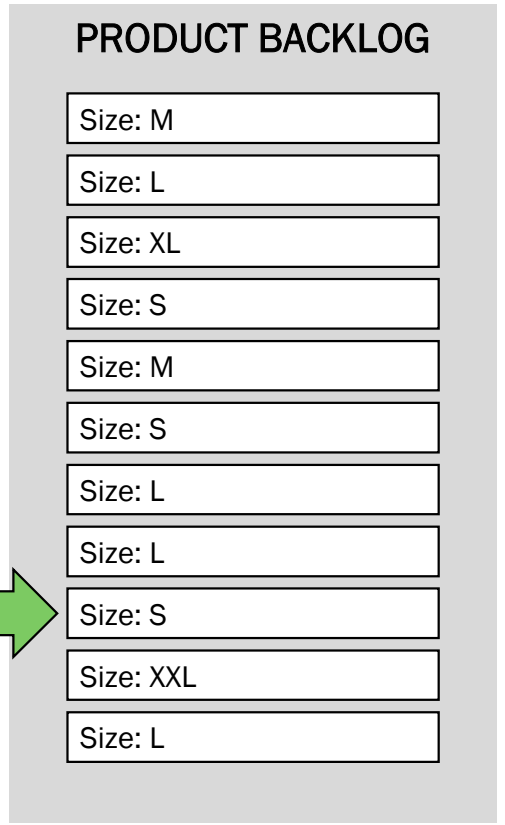
5 minutes

At a Sprint Review one of the stakeholders wants to know when item A is likely to be released.

How would you deal with this question?

- Average Number of PBIs per Sprint = 4
- Sprint Length = 2 weeks




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# Let's Talk about Size and Estimation

## Absolute Estimation

- Traditional, time or effort based
- Least useful for complex work
- Examples are Gantt charts & Function point analysis

## Relative Estimation

- Easy to get started with
- Allows for uncertainty and discussion
- Examples include Story Points, T-shirt sizing and Same-sizing

## Flow Metrics

- Easiest to use
- Data-driven
- Value focused
- Uses throughput and rightsizing

*Many sizing and estimation techniques can be used. Work with the Scrum Team to select a technique that would work best in your environment.*

For more on this topic





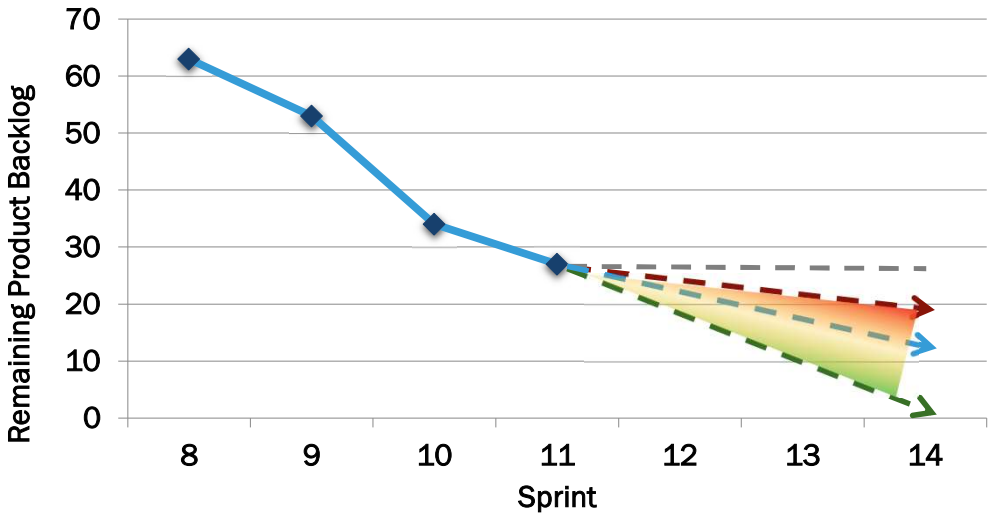

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# A Scrum Team Should Be Able to Measure Progress Against Goals

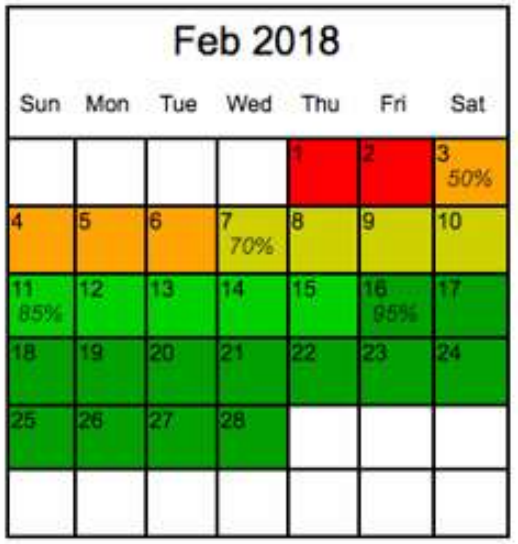



Different ways for visualizing progress can be used. Some are better than others. Consider forecasting the number of items, story points, or other techniques.

Cone of Uncertainty
(Release burndown chart)




Probabilistic Forecasting
(Monte Carlo Simulation)






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## An Agile Solution for Fixed Price, Fixed Date Work

Add Contract Provisions:

- Any requirement that hasn't already been worked on can be swapped out for another of equal size
- Order of requirements can be changed
- Customer may request additional releases at any time at prevailing time and material fees
- Customer may terminate contract early if value has been satisfied for 20% of remaining unbilled contract value


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# TAKE AWAY

**Product Delivery with Scrum**

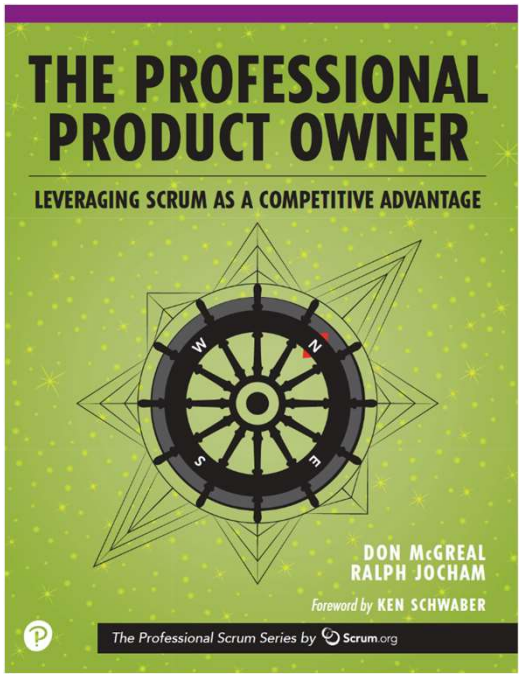
- Product Backlog holds all the work for the Product.
- Product Backlog gives transparency.
- Product Backlog is a living artifact.
- Product Backlog holds all information needed for forecasting, planning and reporting.

Scrum.org/PSM

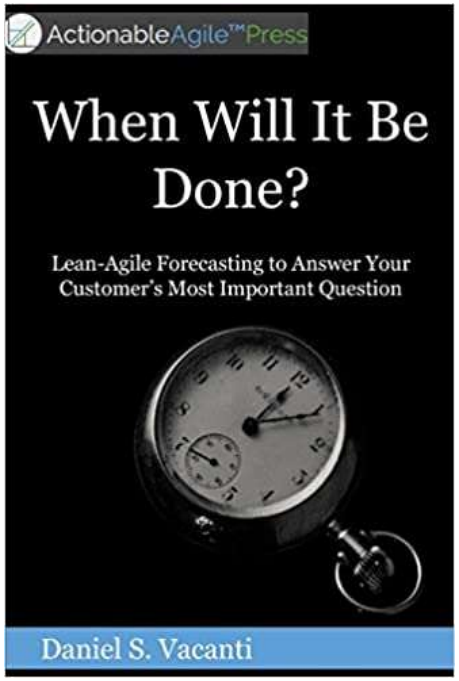
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# Suggested Reading

"The Professional Product Owner" (Don McGreal, Ralph Jocham)

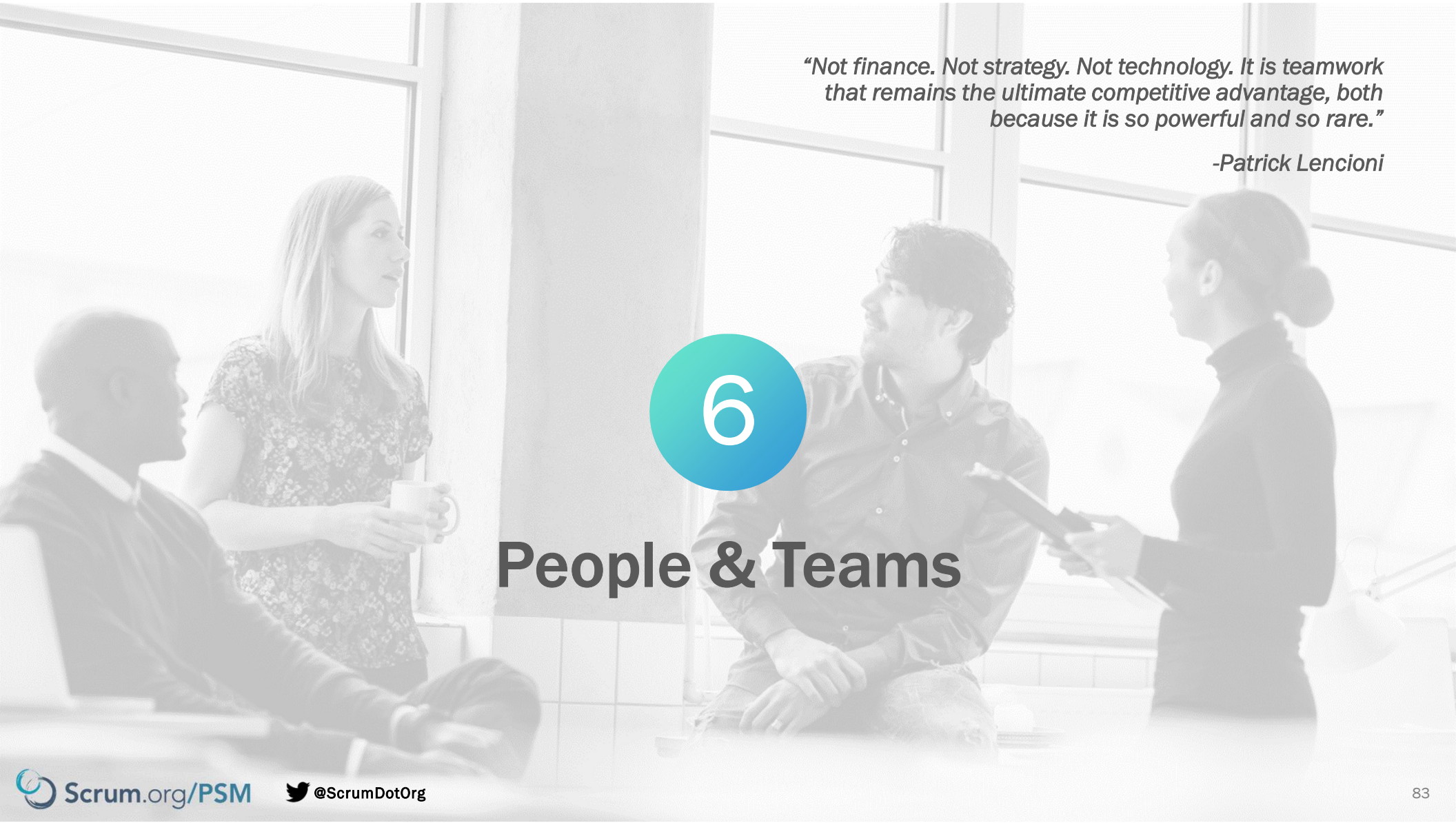



"When It Will Be Done" (Daniel Vacanti)





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*"Not finance. Not strategy. Not technology. It is teamwork that remains the ultimate competitive advantage, both because it is so powerful and so rare."*

*-Patrick Lencioni*

6

# People & Teams

Exercise

# What Scrum Requires




- Mark each statement True or False
- Explain

| | True | False |
|---|---|---|
| Scrum Teams must be co-located | | |
| Two people cannot form a Scrum Team | | |
| Scrum Teams cannot be bigger than 10 members | | |
| A Developer cannot be a Product Owner or a Scrum Master | | |
| Every Developer must be able to perform every type of task | | |
| If Scrum Teams consult external people, they are not self-managing | | |
| All members of the Scrum Team need to be present on the team full-time | | |
| Developers must have clear sub-roles (coder, tester, analyst, writer, …) and accountabilities | | |

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Think of a time you were part of a great team.

What did you appreciate about the experience?

What were the behaviors and characteristics of the team?

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## What Truly Motivates People

External rewards like money (carrot-and-stick) work only for simple, mechanical work

- It has opposite effects in cognitive, complex or creative work

Money counts, but the secret to commitment lies beyond it, in:

- Autonomy – managing my own work
- Mastery – becoming better at my work
- Purpose – making a contribution





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Exercise

# Constructing the Teams

PURPOSE

Explore Scrum Master's contribution to teams coming into existence

5 minutes

Scrum.org/PSM

Your organization is starting the development of a new product line. All 200 people that will be part of the teams have been made available. These people have all required technical and development expertise.

Management asks you, as Scrum expert, to divide them into Scrum Teams.

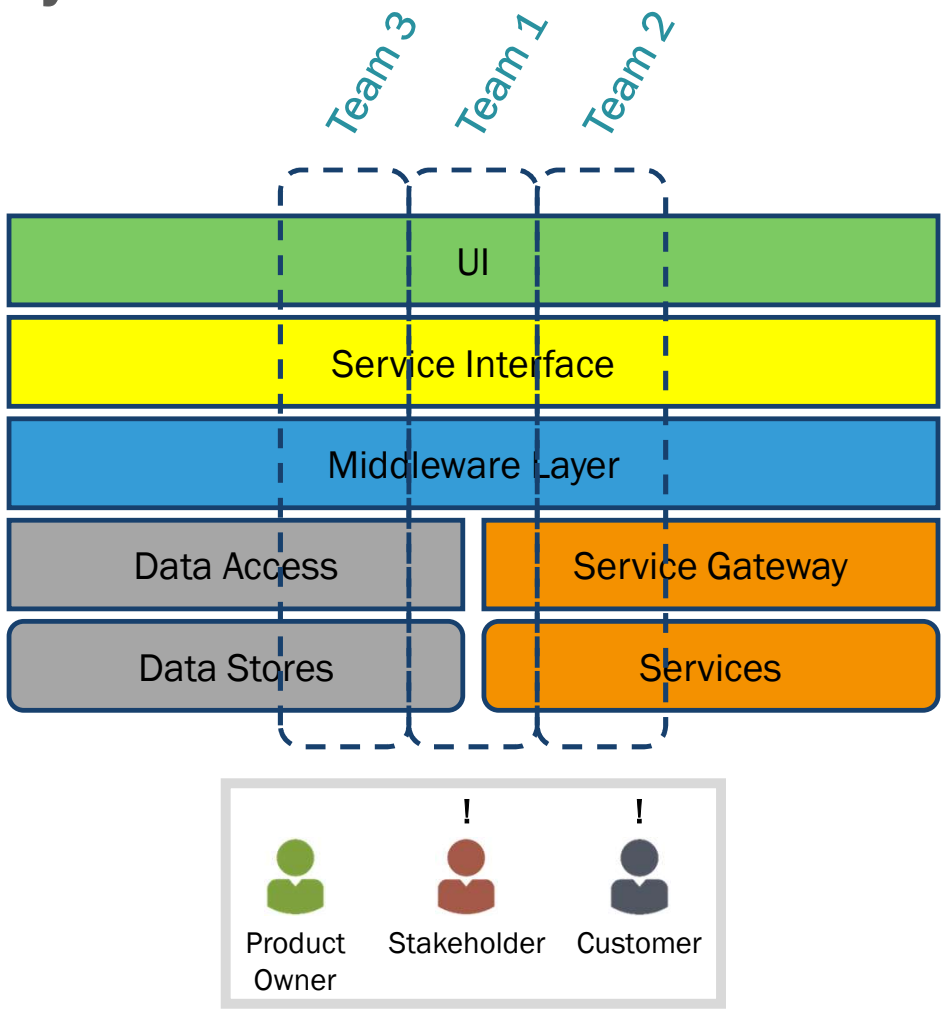
Question: What will you take into account? How will you proceed?

*More on this subject: How Self-Selection Lets People Excel by Sandi Mamoli and David Mole*

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# Feature Teams Enhance Transparency

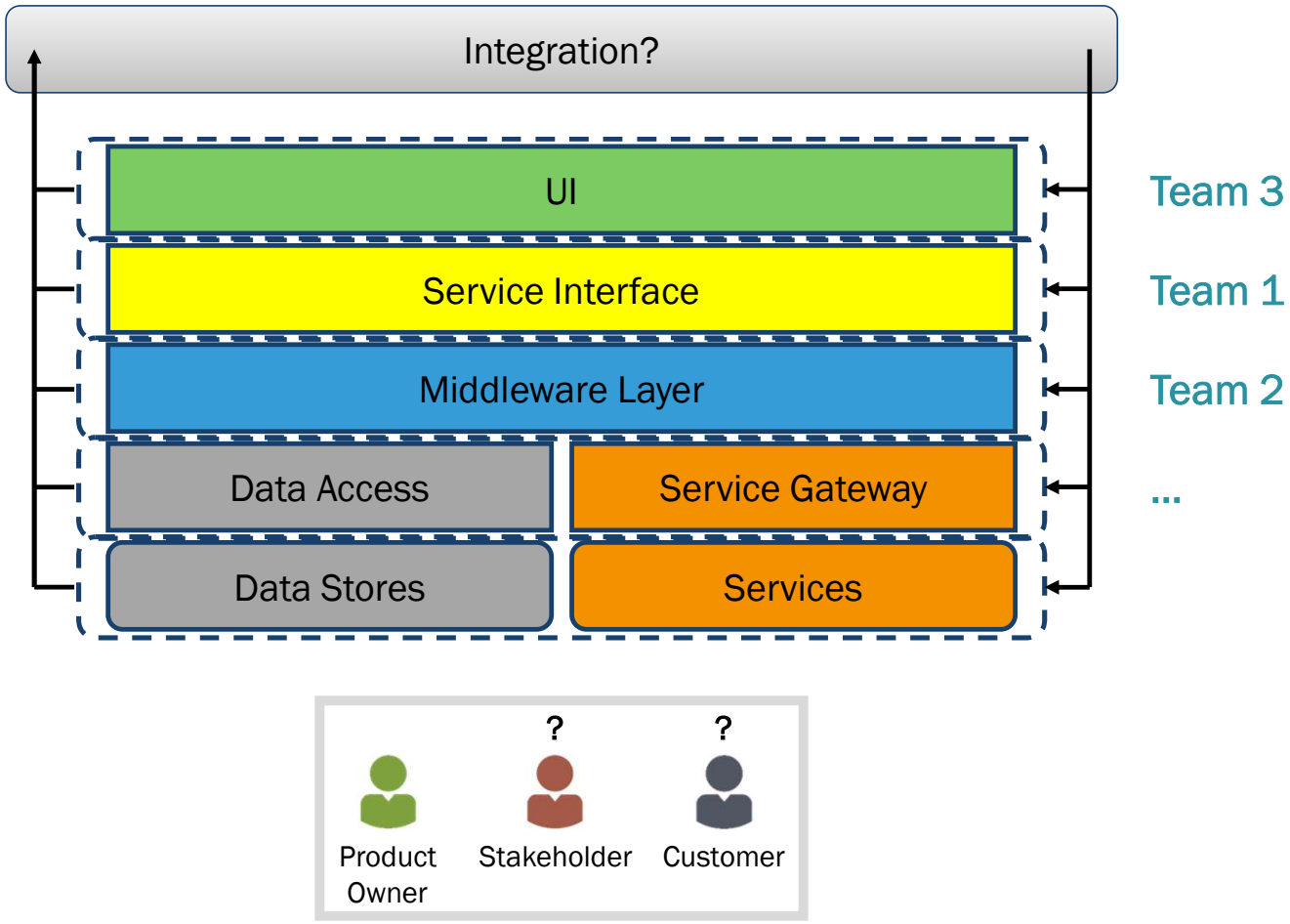
- Each team has all skills to turn Product Backlog into releasable Increments.
- Vertical slicing; work is divided by end-user functionality.
- Work is integrated continuously within each Sprint.
- Transparency ensured; no unknown, undone work.





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# Component or Layer Teams Face Additional Complexities





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Exercise

# The Cindy Problem

PURPOSE

How to deal with scarce skills

5 minutes

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You are Scrum Master for three Scrum Teams. They work from the same Product Backlog, have the same Product Owner, and share a common solution base.

Developers from all three Scrum Teams report that in the next three Sprints they will all be working in one specialist area of the solution. Cindy is the only expert that knows that section well. The teams will need Cindy full-time for their Sprints.

## Question: What do you suggest?



Team 1 Team 2 Team 3 Cindy

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# Scrum Thrives on Self-Managing Teams

- Manager-led work limits agility and other benefits of Scrum.
- Constraints are often set by the organization.
- Scrum provides boundaries and accountabilities for self-management to be more effective.
- Self-management works better against goals.
- Benefits of Scrum increase as self-managed Scrum Teams mature.




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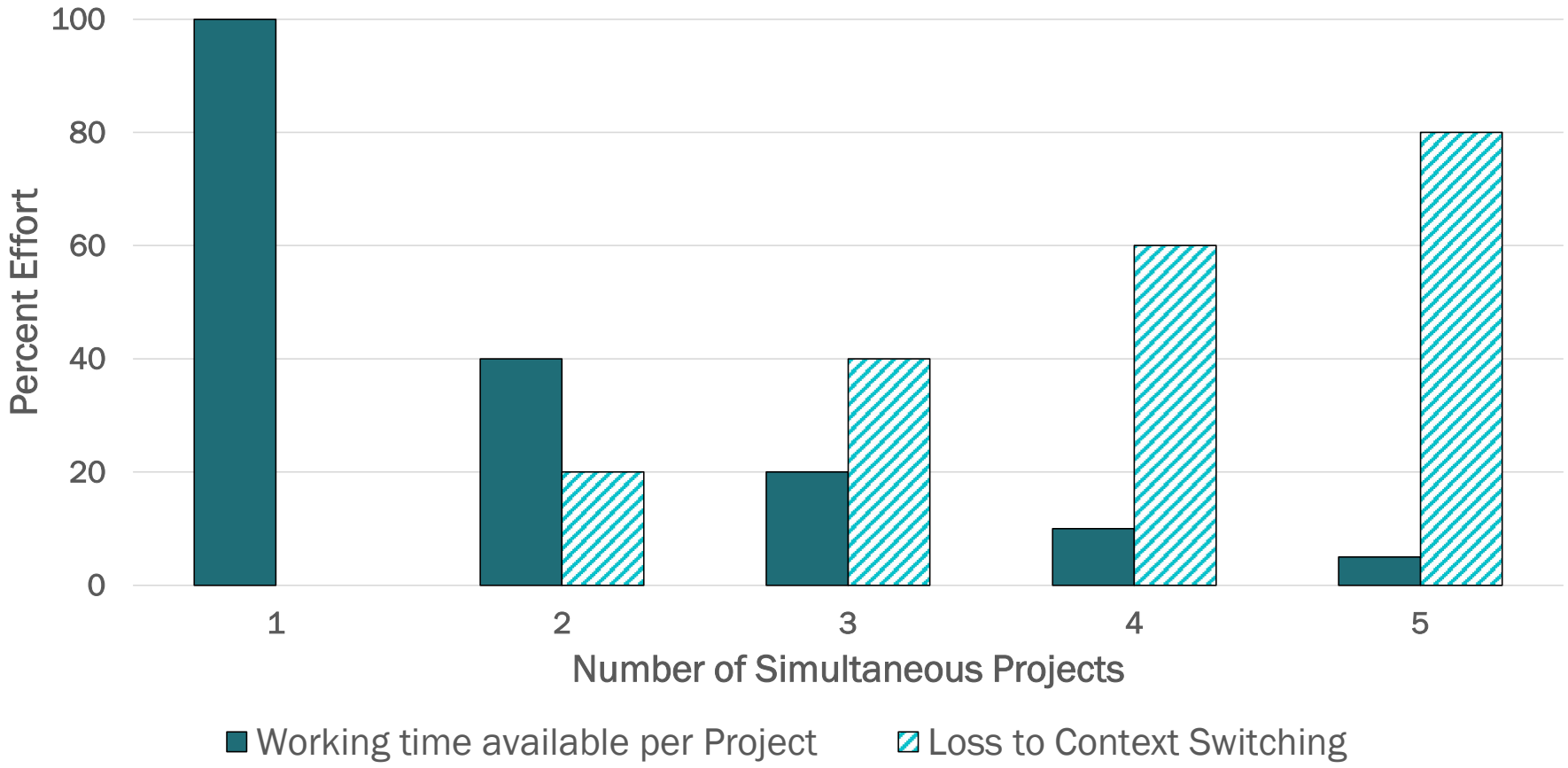



During team formation and start-up, you discover that the Scrum Team has to keep working on other projects to get them done in time.

## What would you advise in this situation? Why?

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# Task Switching



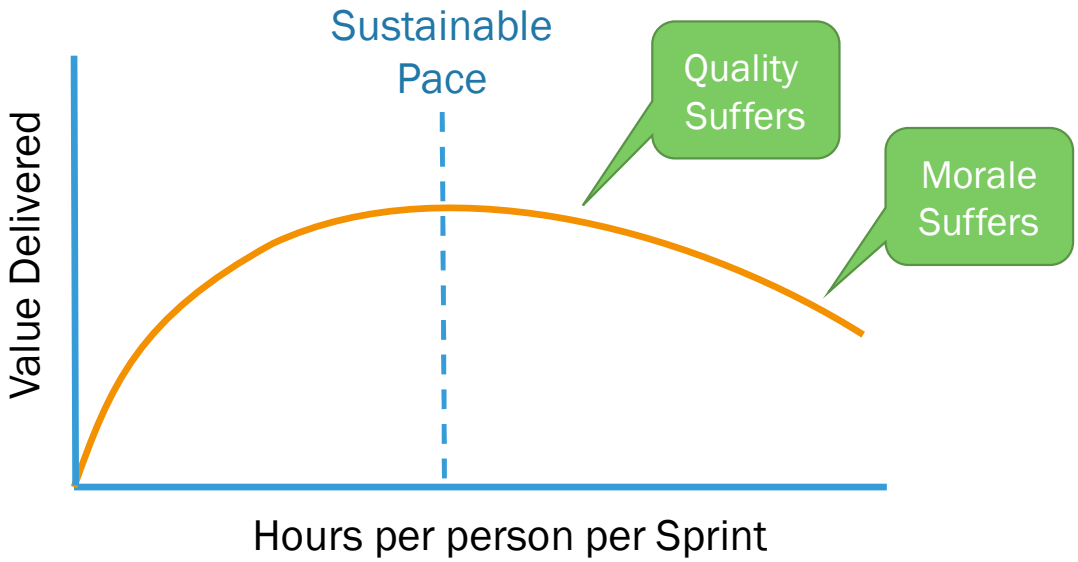



Source: Gerald Weinberg, Quality Software Management: Systems Thinking

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# Sustainable Pace

- People work at a sustainable pace
- If Developers have to consistently work more, quality and creativity drop.





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# Scrum Values

- The Scrum values are the foundation for behavior and practices in Scrum.
- They are closely related to the theory and first principles of Scrum and support teams in their work.
- Scrum Masters can always fall back on these essentials.




Scrum values are the life blood of the Scrum framework.

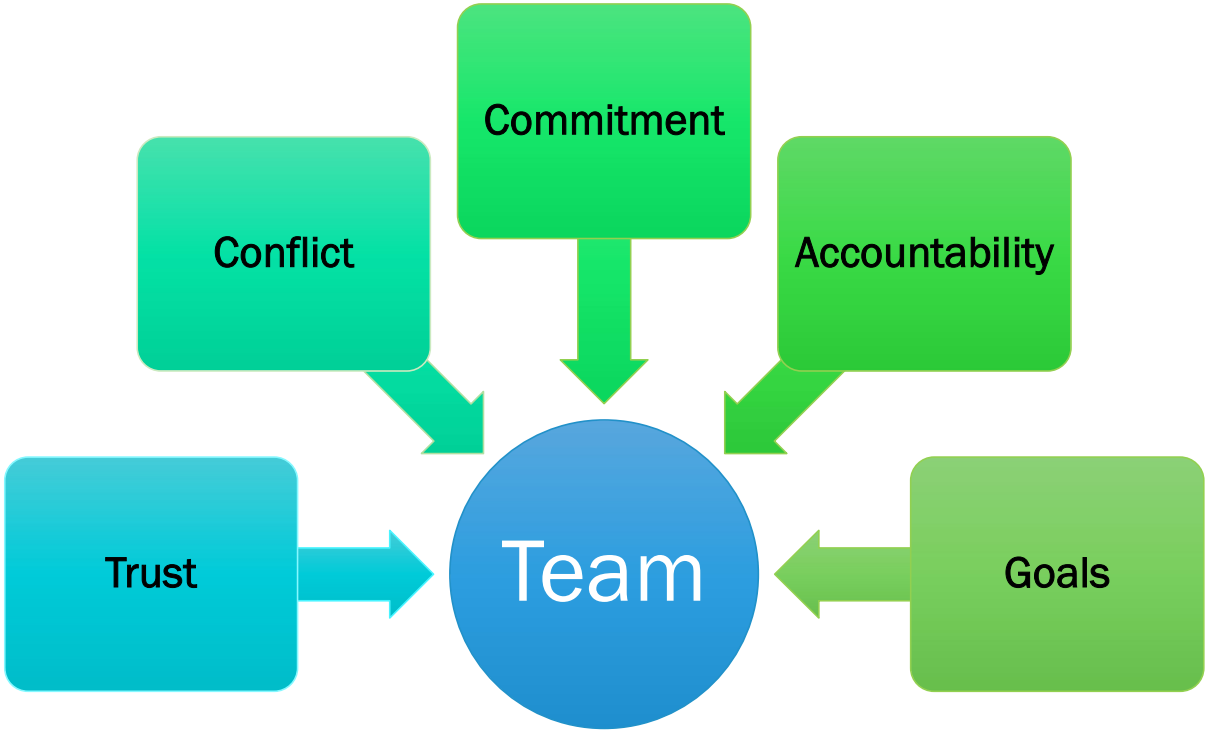

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Assets can turn into dysfunctions and grind a team.

A team requires nurturing, cherishing and attention to avoid team atrophy.

Scrum.org/PSM

# The Assets of a Collaborative Team



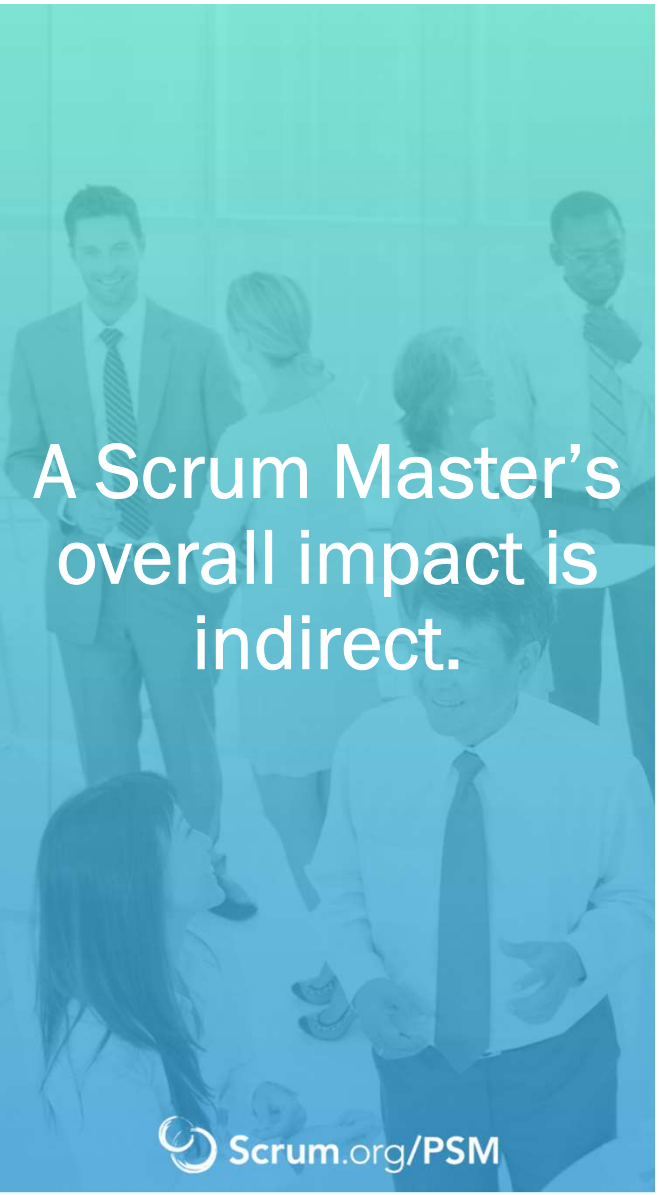

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How does a Scrum Master help a team become collaborative and effective?

How does a Scrum Master help a team stay healthy?

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# A Scrum Master Serves the Scrum Team

- **Lead by example.** Be the first one to be vulnerable. Be a living demonstration of team assets and Scrum values. Admit your missteps.
- **Create an environment of safety.** Encourage debate, support it and keep it productive. Use coaching techniques like open questions.
- **Facilitate decisions.** Try to have key decisions made clear at the end of team discussions, making responsibility and deadlines clear.
- **Learn to read the room.** Be connected without being present.
- **Show patience.** Be okay with silence. Let the team take action.
- **Restrain from solving.** Reveal, not resolve. Be careful not to steer the team towards premature resolution of conflict to protect people. Help team members develop conflict resolution skills.
- **Be comfortable with failure.** Team decisions may not lead to the anticipated outcome. This is part of learning and growth.
- **Care for people.** Listen to them without judgment. Assume positive intent. Meet them where they are and help them find the next step.
- **Show low tolerance for organizational impediments.**

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# A Mindset and Behavioral Shift for Management

## PREDICTIVE MANAGEMENT

- Long-term detailed plans
- Assign and control the work
- Maximize capacity and effort
- Keep all on schedule
- Driven by meetings and reports
- Intervene to fix all problems
- Provide external motivators ($, job title)

## EMPIRICAL MANAGEMENT

- Goals, vision, direction
- Foster the environment
- Help remove impediments
- Attend Sprint Reviews
- Share incremental feedback
- Manage for value
- Autonomy, mastery, purpose

Are you going to be impacted by the change, or are you going to help lead the change?

For more on this topic 




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# TAKE AWAY

## People & Teams

- People take commitments they make more seriously than commitments others make for them.
- Teams are more productive than the same number of individuals.
- Teams and people do their best work when not interrupted.
- Products are more robust when a team has all of the cross-functional skills to do the work.
- Under pressure to “work harder,” quality is automatically and increasingly reduced.
- Changes in team composition often lower productivity for a time.

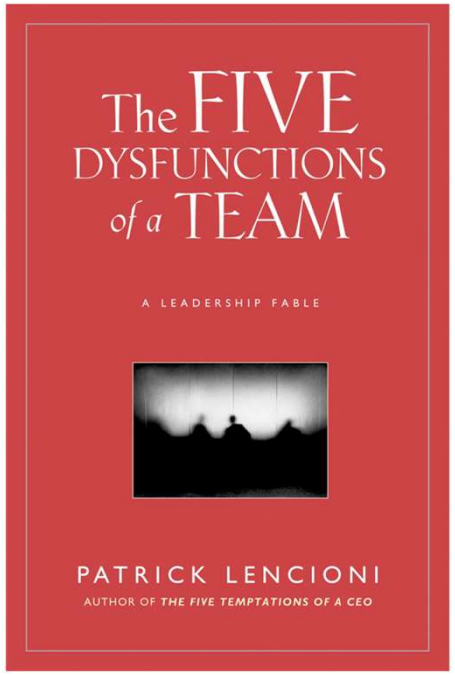



Scrum.org/PSM

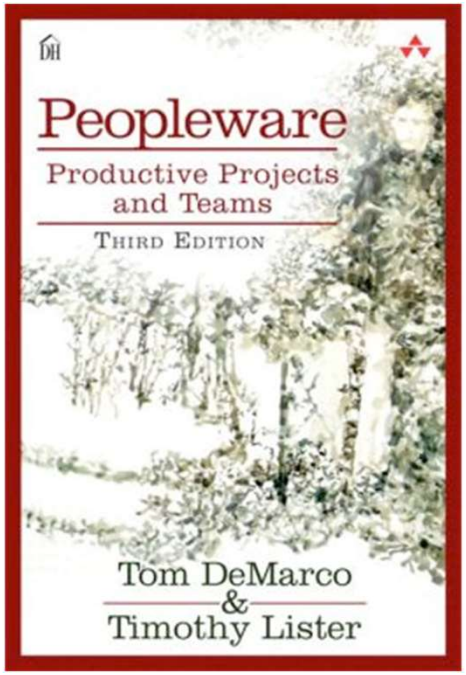
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# Suggested Reading

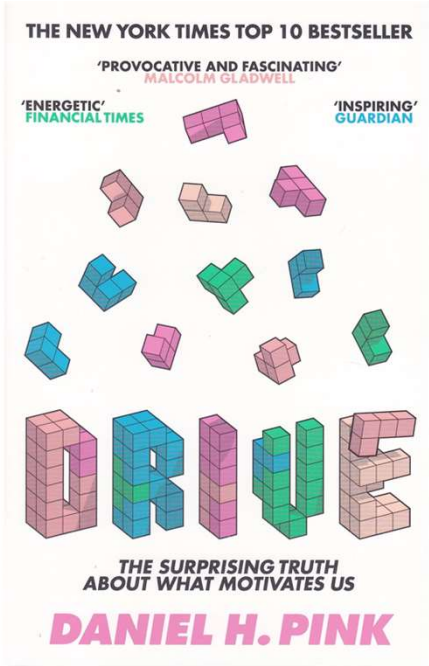
"The 5 Dysfunctions of a Team" (Patrick Lencioni)




"Peopleware" (Tom DeMarco, Timothy Lister)




"Drive" (Daniel Pink)






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*"It does not take a majority to prevail... but rather an irate, tireless minority, keen on setting brushfires of freedom in the minds of men."*

*- Samuel Adams*



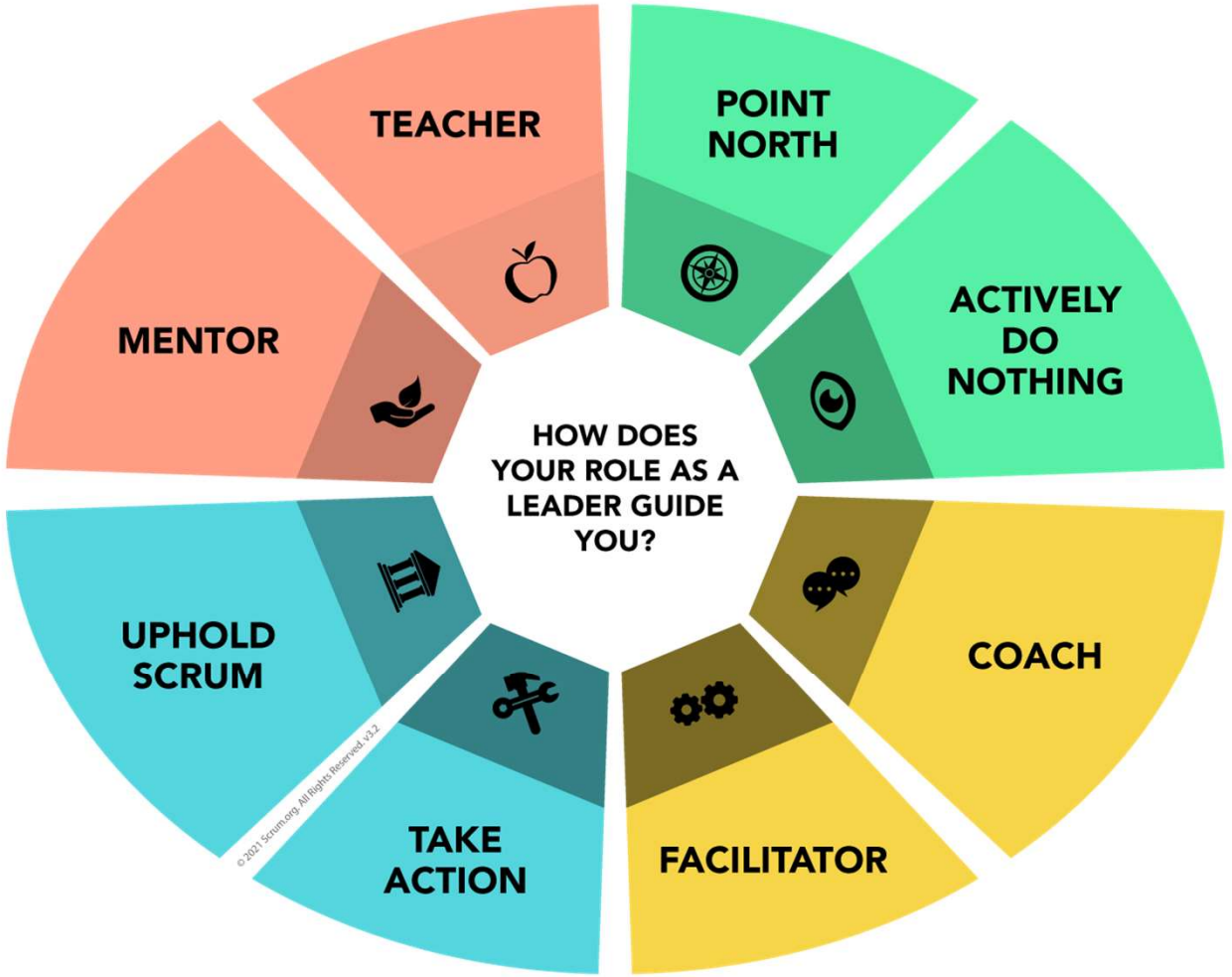
7

# The Scrum Master

There is no methodology for a Scrum Master to follow, yet there are intervention choices and behavioral stances from which to choose depending on context.

Scrum.org/PSM

## Scrum Master Choices




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The Scrum Master's approach will vary based on context. What might a Scrum Master consider?

How will you approach the situation?

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## Scrum Master Responsibilities

- Ensures Scrum is understood and enacted.
- Facilitates Scrum events as needed or requested.
- Helps everyone adhere to Scrum's theory, practices, and rules.
- Helps people embrace and live the Scrum values.
- True leader for the Scrum Team.
- Causes change that improves quality or productivity.
- Embody agility to the organization.

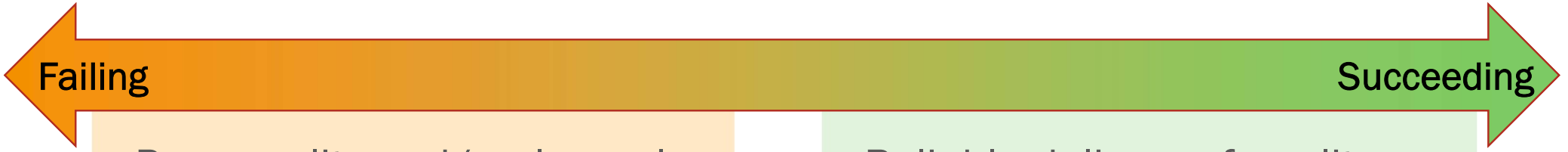
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# What Does it Mean for a Scrum Master to Be a Leader?

| MOVE AWAY FROM | | MOVE TOWARD |
|---|---|---|
| Coordinating individuals and individual contributions | → | Coaching people in Scrum and positive team behavior by embodying the Scrum values |
| Providing answers as a subject-matter expert | → | Enabling self-management within Scrum Teams |
| Investing in specific outcomes (budget and scope) | → | Helping Product Owners manage Product Backlogs and work with stakeholders |
| Deadlines | → | Focusing Product Owners on flow and value |
| Prescribing technical solutions | → | Helping Developers effectively use and improve the Definition of Done |
| Fixing problems | → | Guiding Scrum Teams to discover what works best for them |


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# Measuring the Success of a Scrum Master

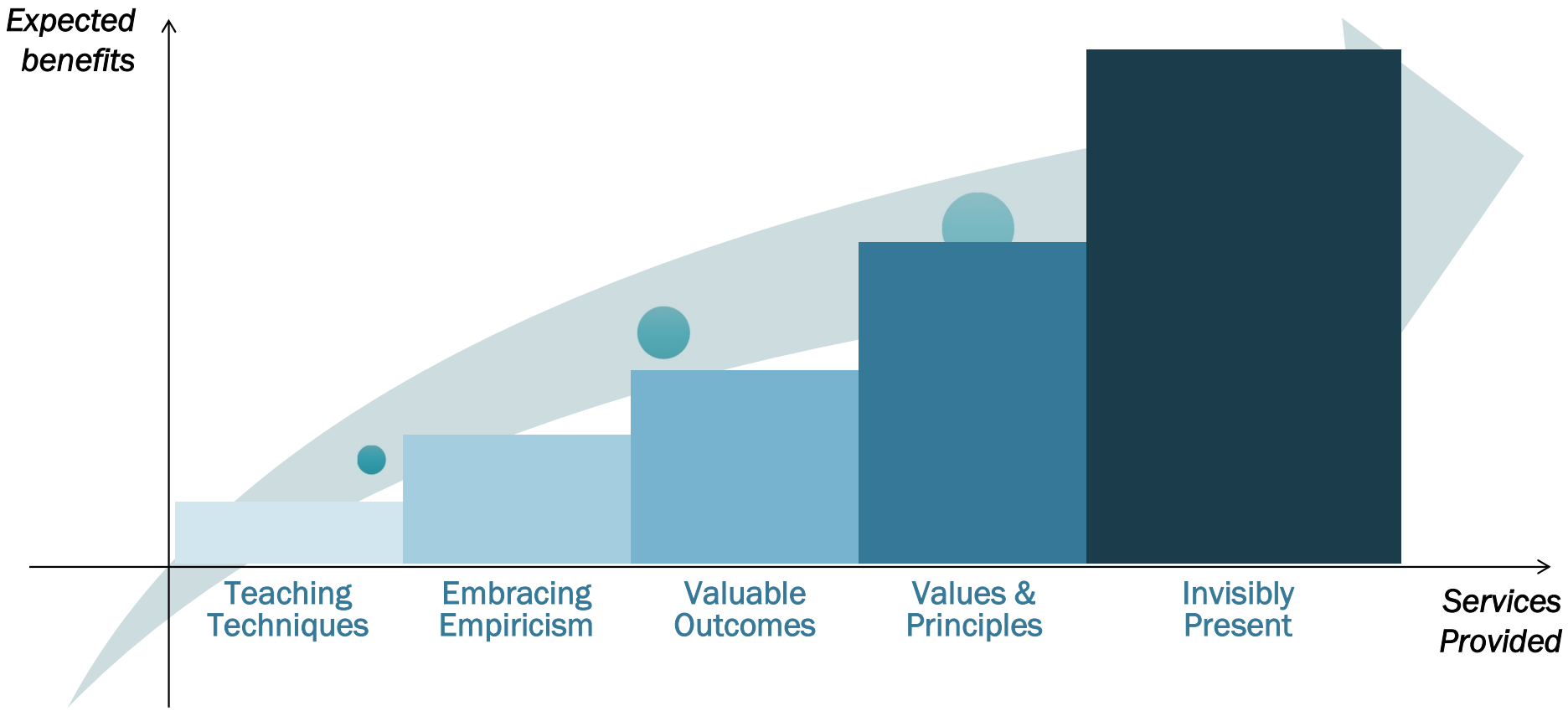



- Poor quality and/ or low value and inconsistent delivery
- Inconsistent or mechanical Scrum
- Low morale
- Stagnation or degradation
- Dependency on Scrum Master

- Reliable delivery of quality, valuable Increments
- Solid understanding of Scrum framework, theory, and values
- High morale
- Continuous improvement and learning
- Self-sustaining


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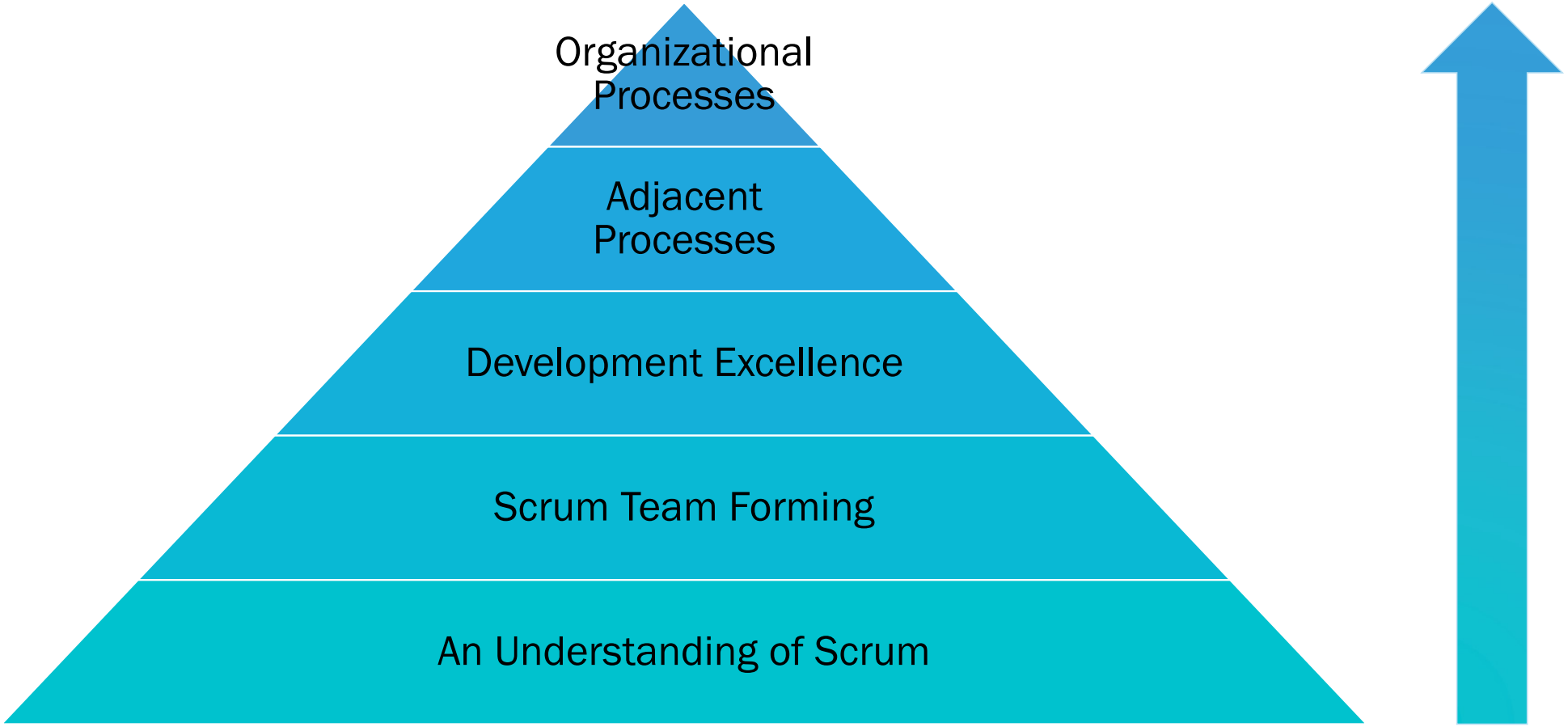
# A Scrum Master Provides Services





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# A Scrum Master Removes Impediments




Source: Dominik Maximini


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Exercise

# Scrum Master Skills




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List the skills and traits a Scrum Master needs to be effective and successful.

| SKILLS | TRAITS |
|---|---|
| | |
| | |
| | |
| | |
| | |

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# TAKE AWAY

**The Scrum Master**

- The Scrum Master's focus is the understanding and effective usage of the Scrum framework.
- The Scrum Master teaches, coaches and mentors the Scrum Team and the organization.
- Being a Scrum Master requires distinct skills.

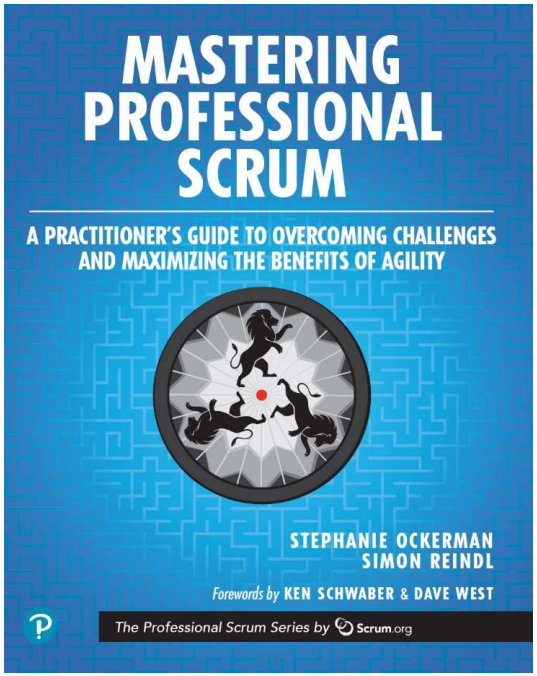




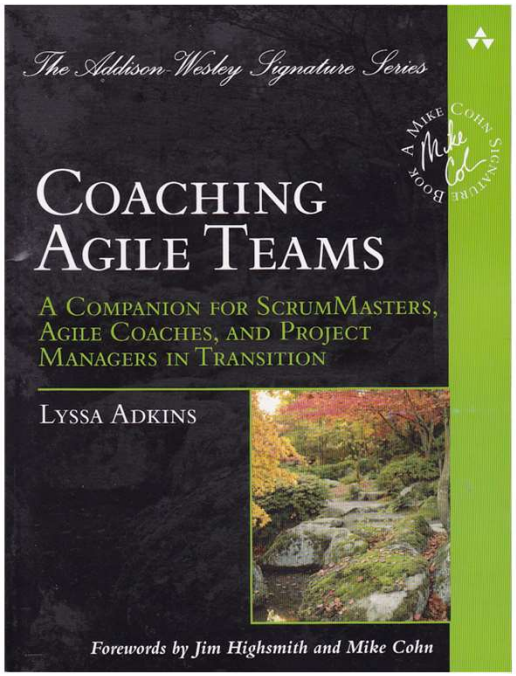

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# Suggested Reading

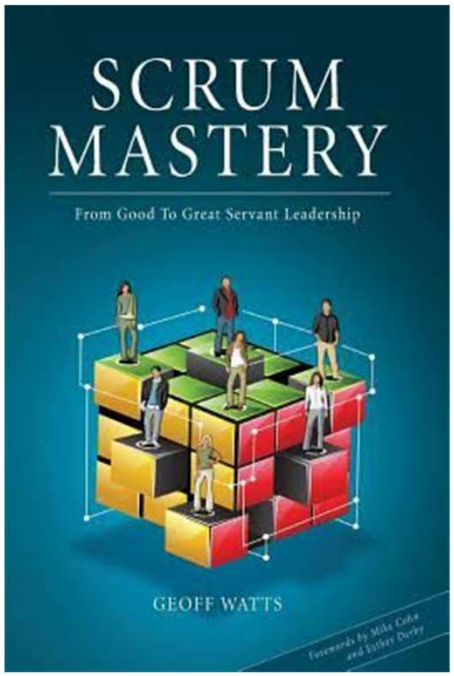
"Mastering Professional Scrum" (Stephanie Ockerman, Simon Reindl)




"Coaching Agile Teams" (Lyssa Adkins)

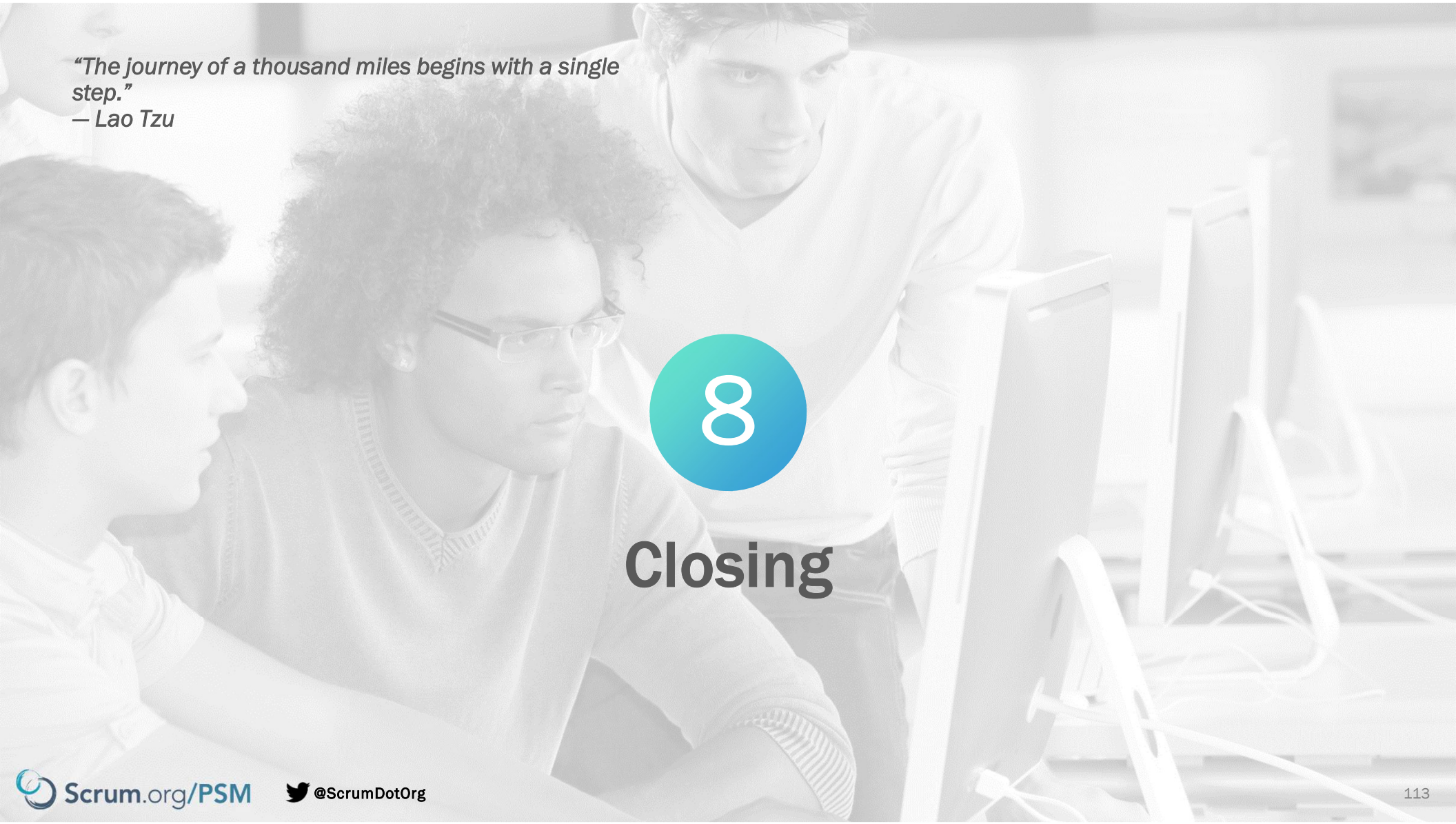



"Scrum Mastery" (Geoff Watts)






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*"The journey of a thousand miles begins with a single step."*
*— Lao Tzu*

8

# Closing

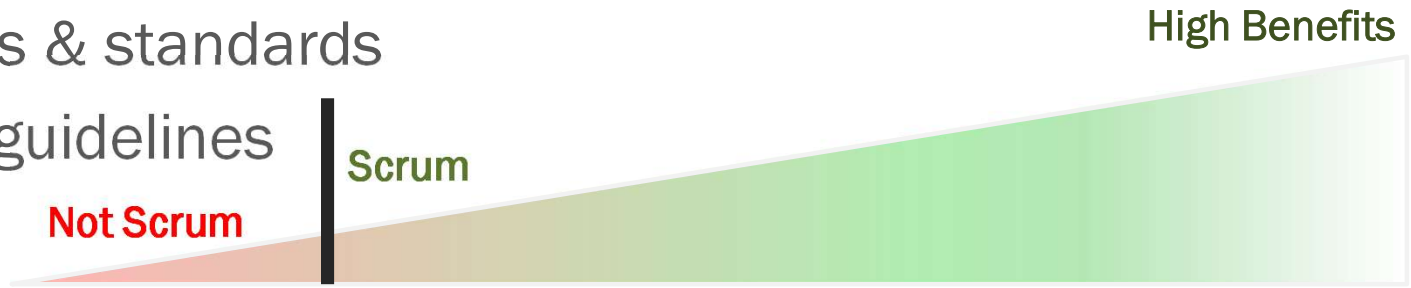
## Hard Choices

- Adopting Scrum requires hard choices.
- Modifying Scrum will not solve the problem, but it may hide it for awhile.
- Changing everything overnight will not solve the problem either.
- Be patient but keep challenging the status-quo.


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# Many Ways to Maximize Scrum

- Team effectiveness through collaboration, autonomy & self-management
- Product Management - Vision, Value, Validation
- Improve flow with Kanban
- Integrate User Experience
- A Definition of Done that reflects usable
- Engineering practices & standards
- Quality standards & guidelines
- Skills (training)




**“ScrumAnd”**


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# Three Things You Wanted to Know (Re-Visit)

- Did we cover what you absolutely wanted to know?
- Did we set some questions aside that we still need to go into?




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I've had 2 great days of discovery about being a Professional Scrum Master. But when I go back to work, I still have to deal with many old ways of working (dates, actuals, predictions).

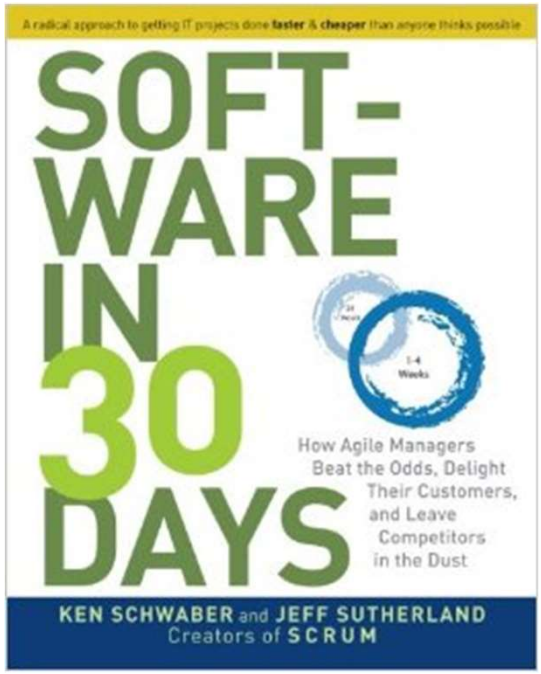
Take 5 minutes to identify 3 actionable ideas or improvements from this class you will try.

Pair up with someone and discuss your respective actionable items. Help each other refining them.

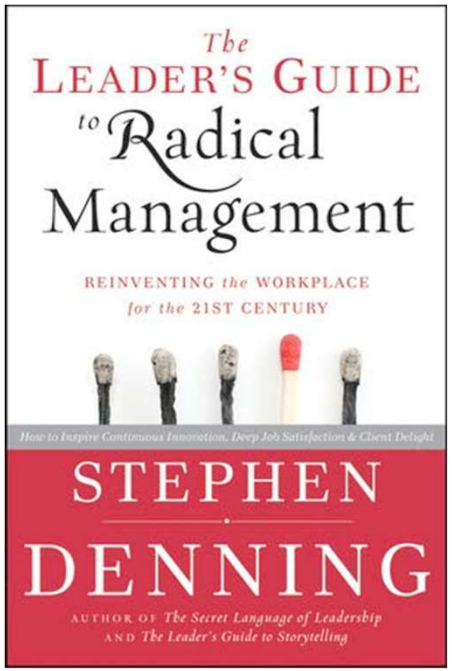
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# Suggested Reading

"Software in 30 Days" (Jeff Sutherland, Ken Schwaber)




"Radical Management" (Stephen Denning)




"Fixing Your Scrum" (Ryan Ripley, Todd Miller)





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# Inspect Your Knowledge – Feedback in 14 Days or Less!




Over the past 2 days, you have learned the importance of inspection, adaptation, and fast feedback cycles. To reinforce those concepts, if you attempt the Professional Scrum Master I (PSM I) certification assessment within 14 days and do not score at least 85%, you will be granted a 2nd attempt at no further cost.




- Test your basic knowledge of Scrum and learn from immediate feedback by taking an Open assessment: www.scrum.org/open-assessments
- Use the Open assessments to prepare for Level I assessments




- As a student of this course, you are eligible for a $100 discount on the advanced Professional Scrum Master II assessment.
- Email support@scrum.org for a coupon to take PSM II at $150 ($250 retail price).


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# Professional Scrum Competencies

**www.scrum.org/professional-scrum-competencies**

The Professional Scrum Competencies help guide an individual's personal development with Scrum.

Benefit from a common understanding of the competencies and focus areas to evaluate and balance your team's proficiencies based on your unique needs.

See how all Scrum.org courses map to the competencies and focus areas by visiting:

[www.scrum.org/courses/professional-scrum-training-competency-mapping](www.scrum.org/courses/professional-scrum-training-competency-mapping)




**Understanding and Applying the Scrum Framework**

- ✓+ **Empiricism**
- ✓+ **Scrum Values**
- ✓+ **Scrum Team**
- ✓+ **Events**
- ✓+ **Artifacts**
- ✓ **Done**
- Scaling

**Developing People and Teams**

- ✓+ **Self-Managing Teams**
- ✓ **Facilitation**
- ✓ **Leadership Styles**
- ✓ **Coaching and Mentoring**
- Teaching

**Managing Products with Agility**

- ✓ **Forecasting and Release Planning**
- Product Vision
- ✓ **Product Value**
- ✓ **Product Backlog Management**
- Business Strategy
- Stakeholders and Customers

**Developing and Delivering Products Professionally**

- ✓ **Emergent Software Development**
- ✓ **Managing Technical Risk**
- ✓ **Continuous Quality**
- Continuous Integration
- Continuous Delivery
- Optimizing Flow

**Evolving the Agile Organization**

- Organizational Design and Culture
- Portfolio Planning
- Evidence-Based Management™

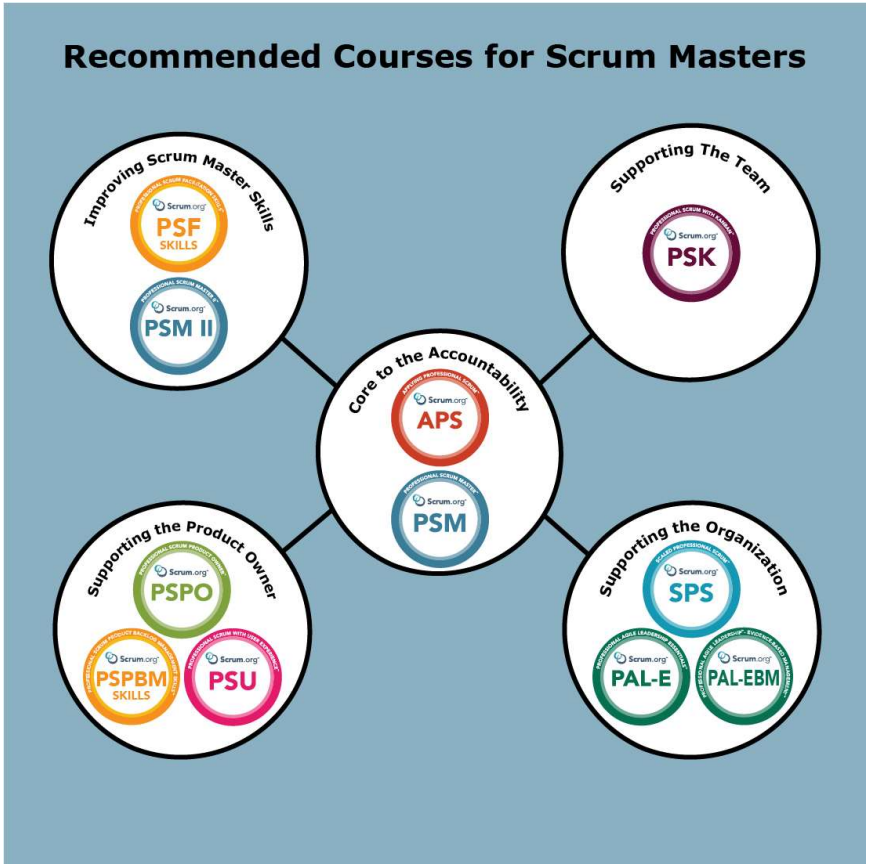
✓ The Focus Area is covered in the class
✓+ The Focus Area has deep coverage in the class


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# Continue With Your Training Path

Additional Training Paths include:

- Product Owner
  www.scrum.org/courses/recommended-courses-product-owners
- Developers
  www.scrum.org/courses/recommended-courses-developers
- Agile Leader
  www.scrum.org/courses/recommended-courses-agile-leaders




www.scrum.org/courses/recommended-courses-scrum-masters


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# Continue Your Learning Online




www.scrum.org/pathway/scrum-master

## Additional Pathways include:

- Product Owner
  www.scrum.org/pathway/product-owner-learning-path
- Developers
  www.scrum.org/pathway/team-member-learning-path
- Agile Leader
  www.scrum.org/pathway/agile-leader-learning-path

## Additional References:

- Kanban Guide for Scrum Teams
  How Kanban and flow-principles can improve your effectiveness as a Scrum Team
  www.scrum.org/resources/kanban-guide-scrum-teams


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# Your Scrum.org Profile



**Reshma (Simran Nagrani**
(her/hers)

Agile For Growth

**Languages**
English

**Country**
India

Class Schedule

**Social Media**

## Reshma (Simran)'s Certifications

- Professional Scrum Trainer
- Professional Scrum Master I
- Professional Scrum Master II
- Professional Scrum Master III
- Professional Scrum Product Owner I
- Professional Scrum Product Owner II
- Professional Scrum Product Owner III
- Professional Scrum Developer I
- Professional Agile Leadership I
- Professional Scrum with User Experience I
- Professional Scrum Facilitation Skills

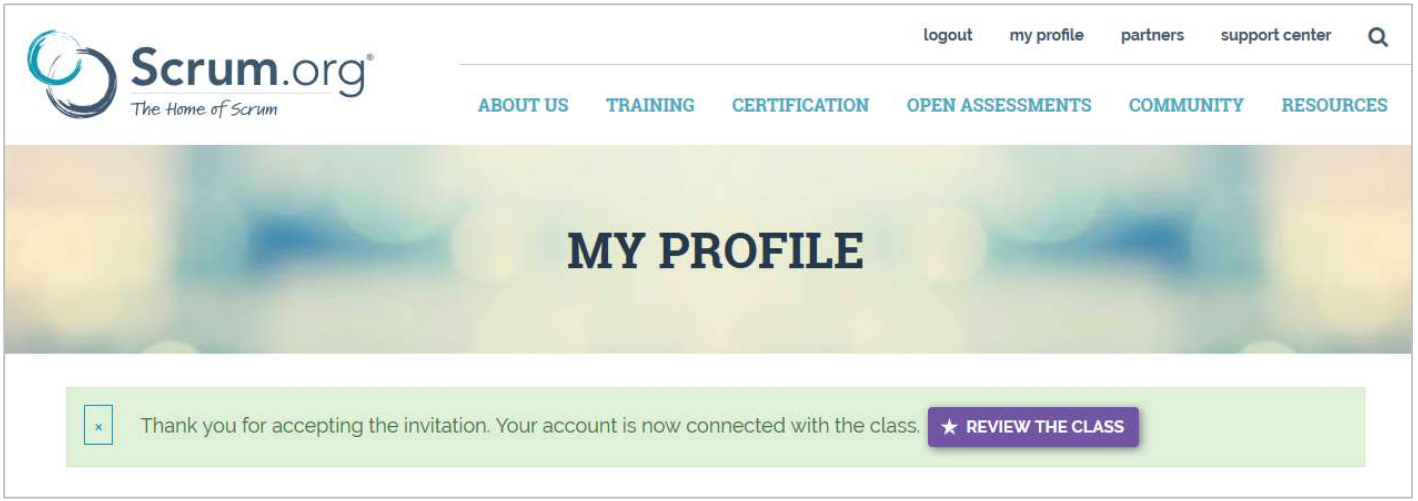
## My Classes

**Professional Scrum Master II**
Jul 6-7, 2019
Review Class

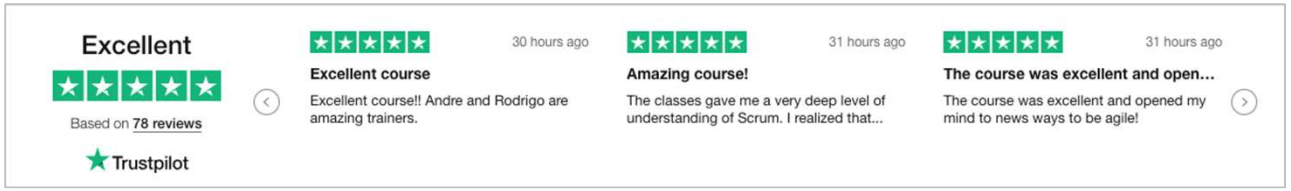
**Professional Scrum with User Experience**
Feb 8-9, 2020
Review Class

# Review Your Class Experience Using Trustpilot

Share your experience with other potential students!




Your review will be visible on our website:





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# Feedback

Feedback is important, and we take it seriously. Your feedback helps us to continually inspect and adapt our courses.

Share your feedback on the class you attended at:

www.scrum.org/feedback


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# Connect with the Scrum.org community



**Forums**
Scrum.org



**Twitter**
@scrumdotorg



**LinkedIn**
Linkedin.com/company/scrum-org



**Facebook**
Facebook.com/scrum.org



**RSS**
Scrum.org/RSS

Scrum.org | Blog
Insights from the Scrum.org Community of Experts

SCRUM PULSE
WEBCAST


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# Thank you!





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